Teaching sand to detect particles

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SUNY Old Westbury
September 30th 2026



Brookhaven
National Laboratory



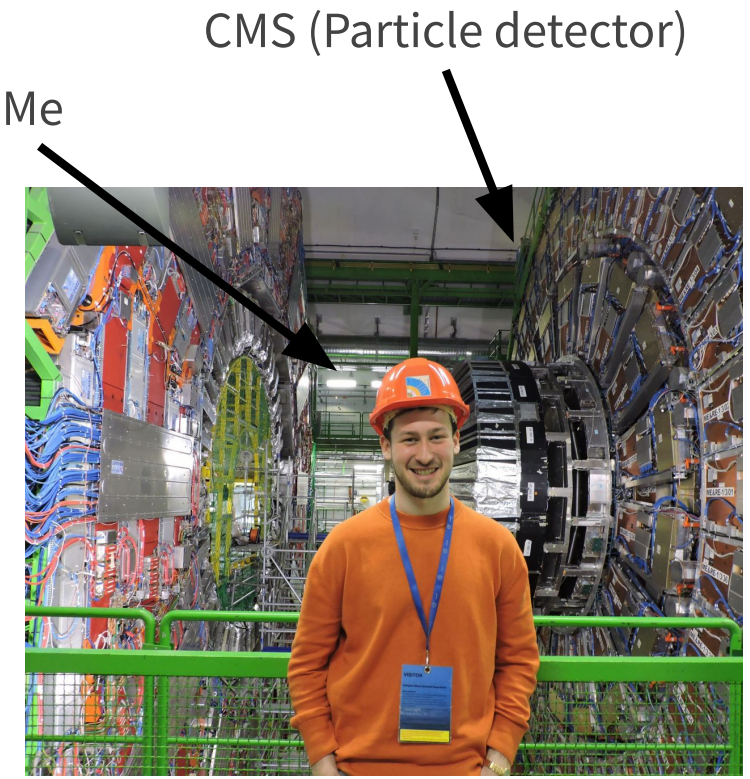
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ATLAS (Another particle detector)

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- I work on **Higgs boson physics** and **tracking detectors**

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- Give you an **introductory idea** of these topics.



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Please raise your hand if you have a question!



Outline

Part I: Why and how we study particle physics

Part II: Detecting particles with silicon

Next section

Brookhaven National Laboratory

Part I: Why and how we study particle physics

Part II: Detecting particles with silicon

Overview

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Brookhaven National Laboratory

Why do we study particle physics?

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→ To understand nature at its most fundamental level: What is the universe made of? How do its constituents interact?

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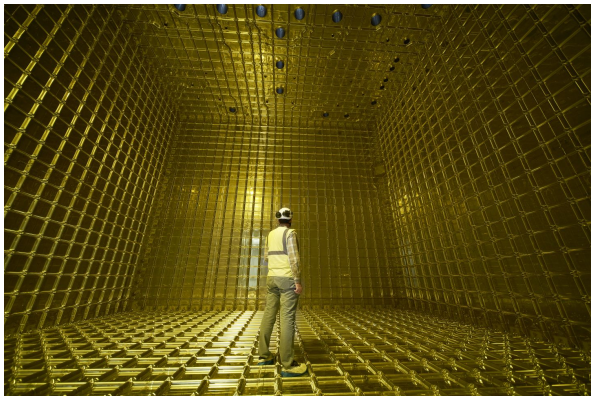
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Intensity frontier



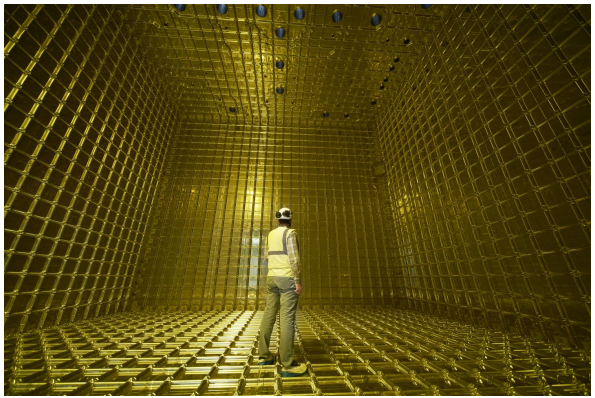
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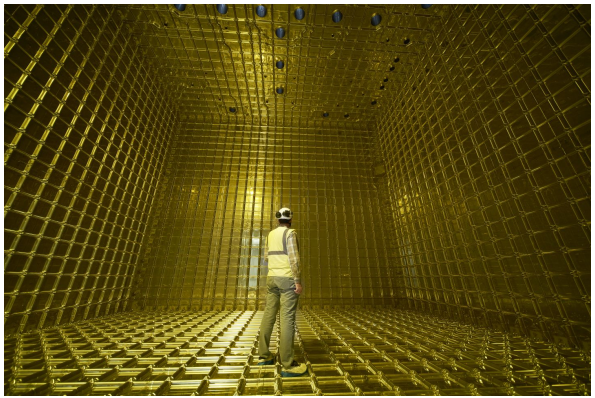
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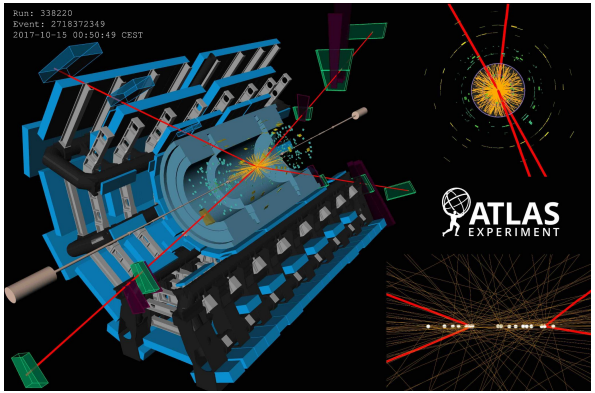
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Energy frontier



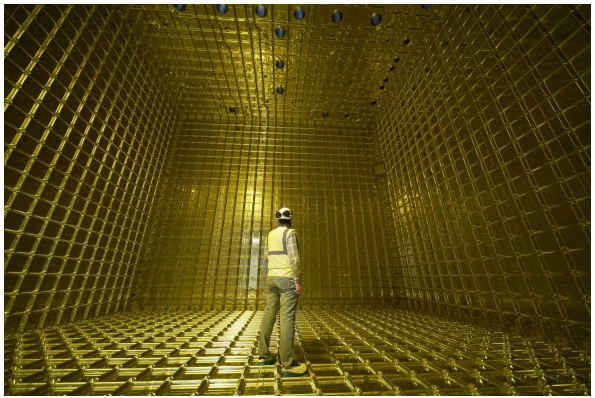
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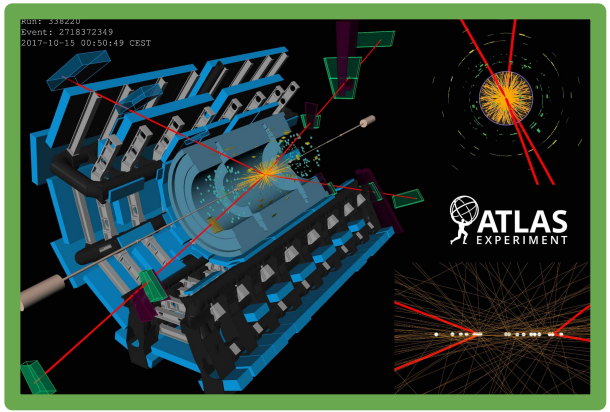


Cosmic frontier



Focus today:

Energy frontier



(Some of) The big physics questions

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Brookhaven National Laboratory

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  - Why is there **more matter** than **antimatter**?

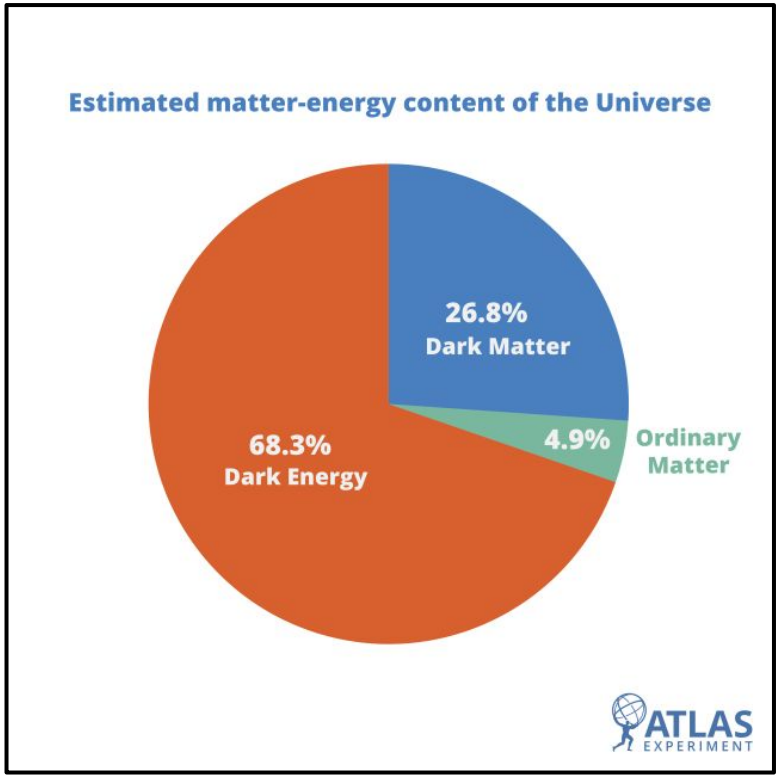
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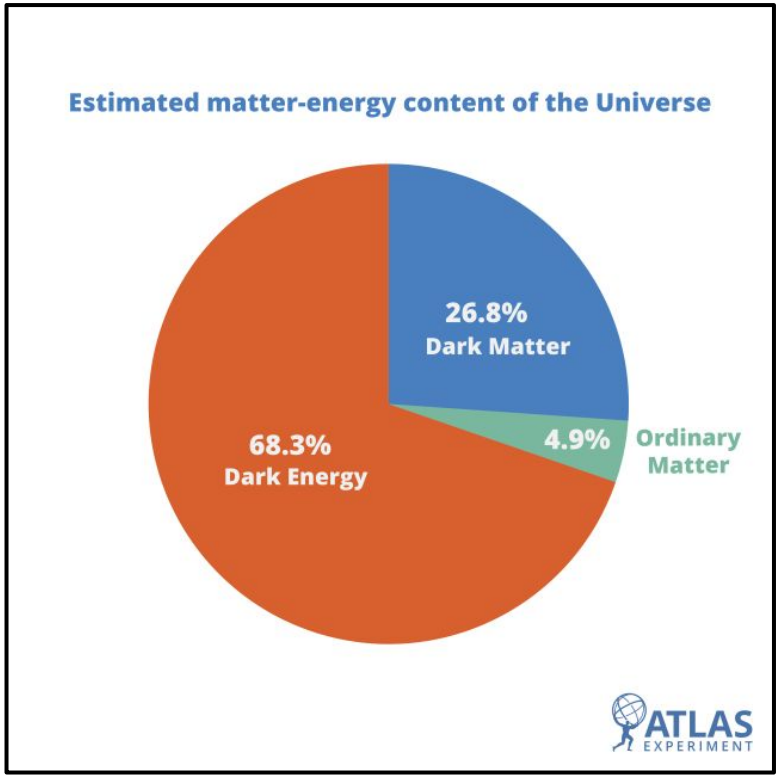
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Searching for Dark Matter with the ATLAS detector

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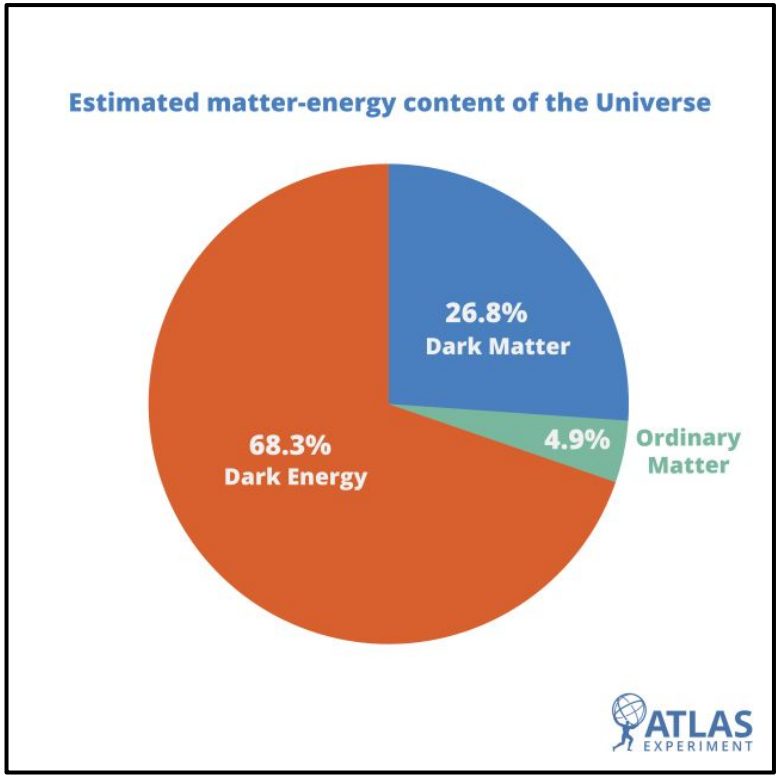


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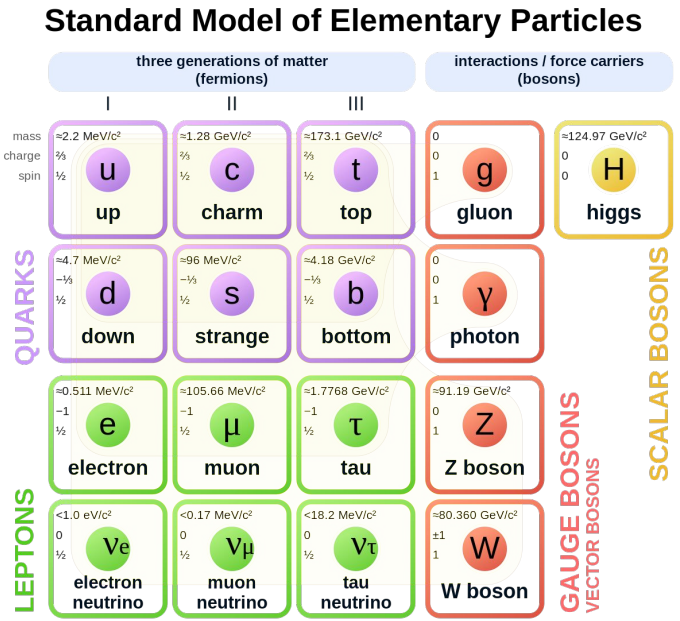
- Different questions tied to different **particles we can study**
  - **For example:** Many of these questions have potential ties to the **Higgs boson**!



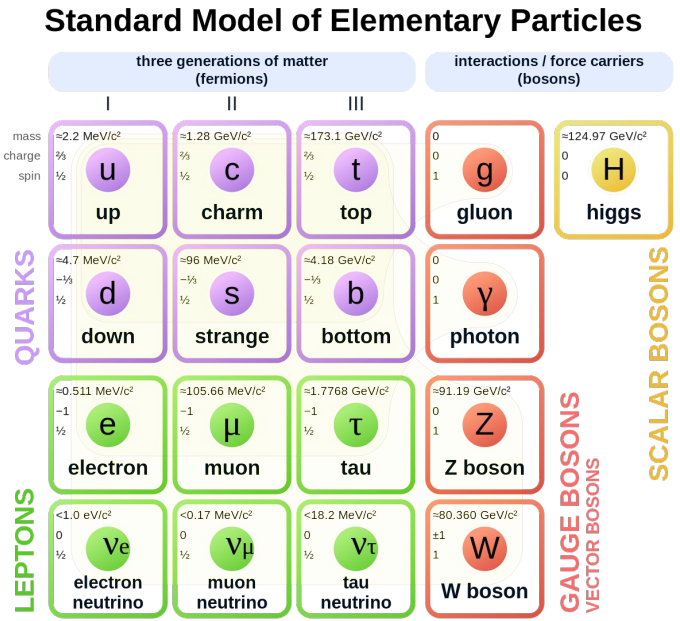
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The standard model

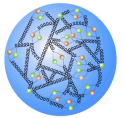
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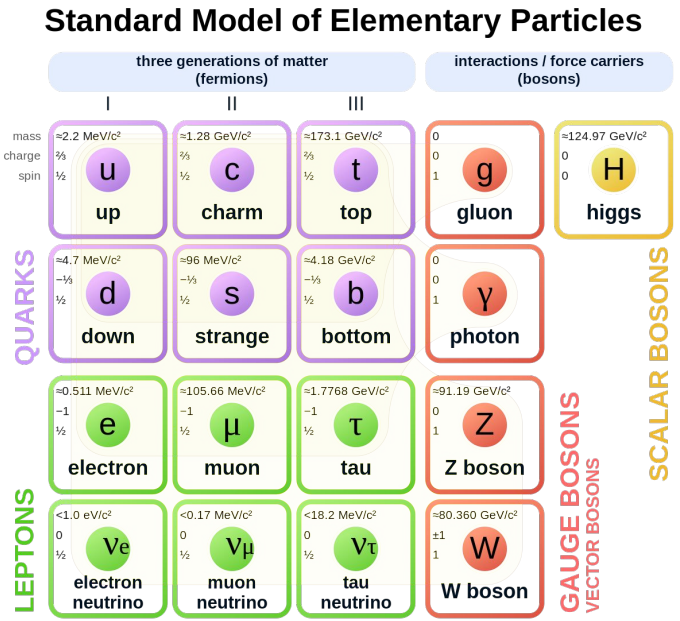


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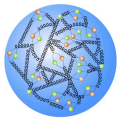


The standard model

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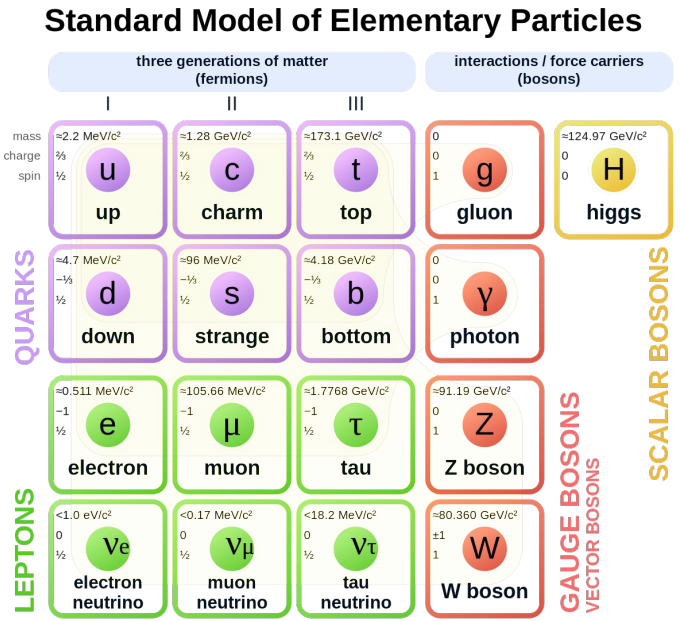
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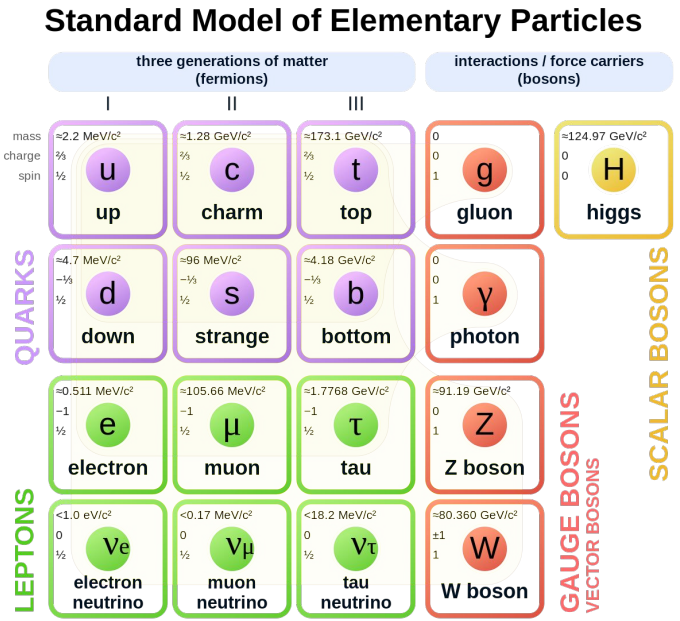


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- Suggests the existence of **BSM (Beyond the Standard Model) physics**





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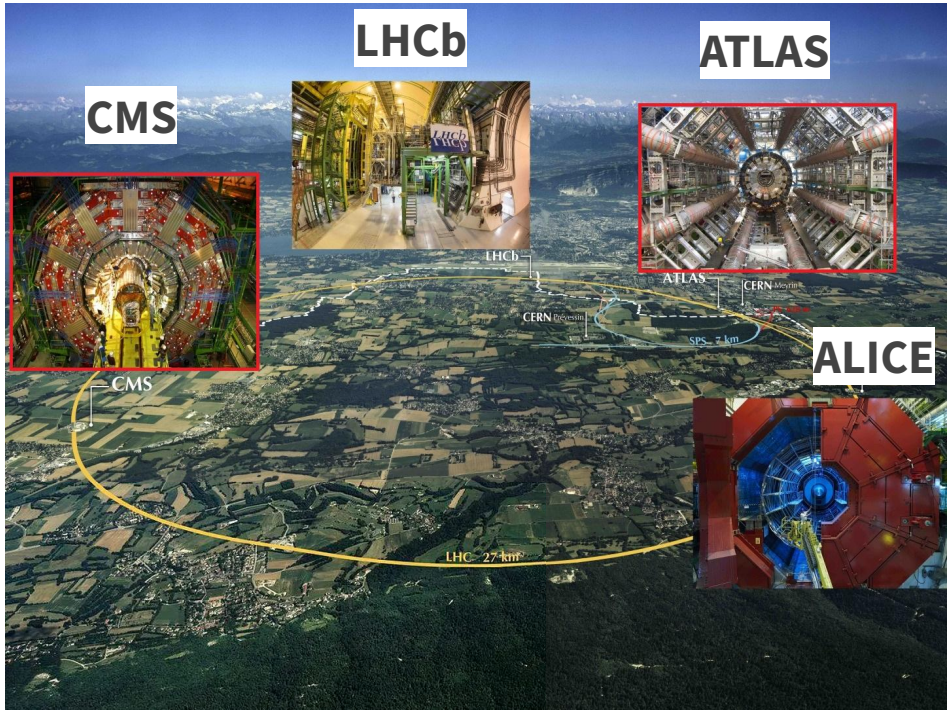


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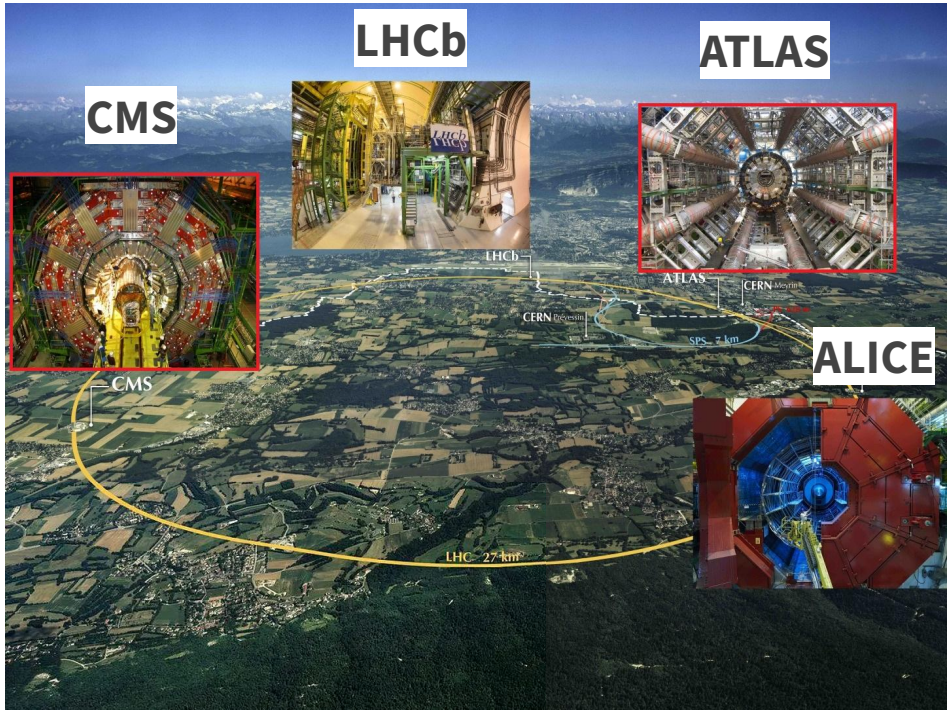
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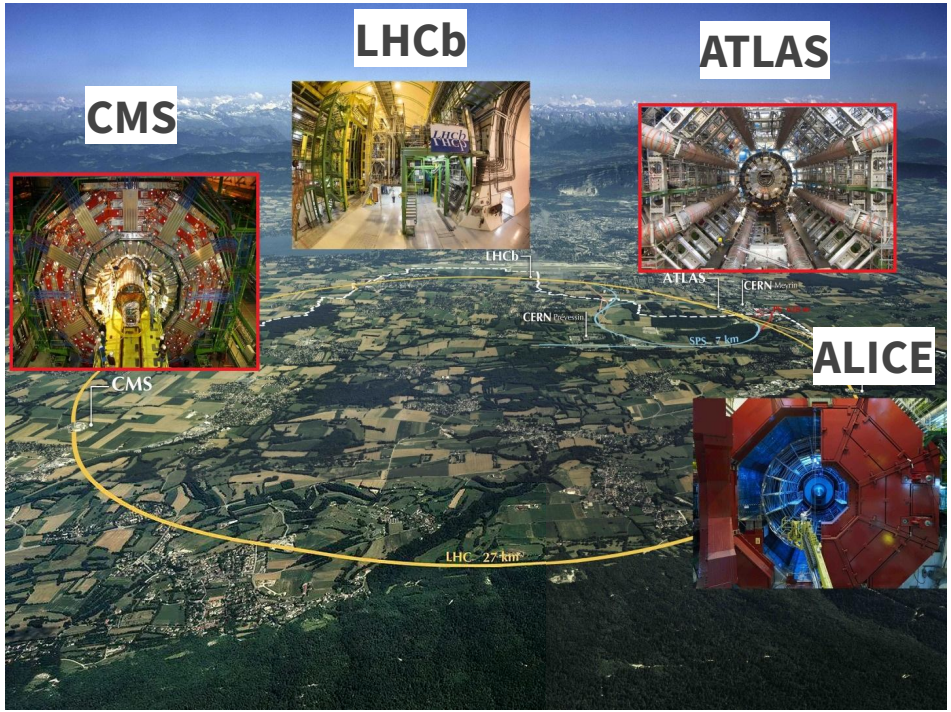
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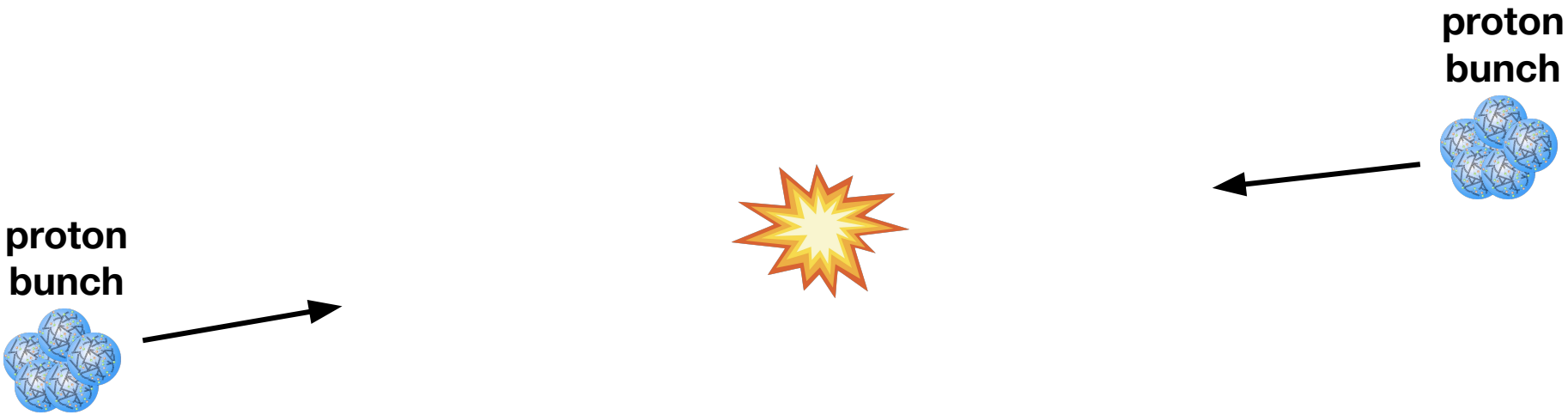
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- The LHC is a **Large** Hadron Collider:
  - **~ 20 minute drive from ATLAS to CMS!**



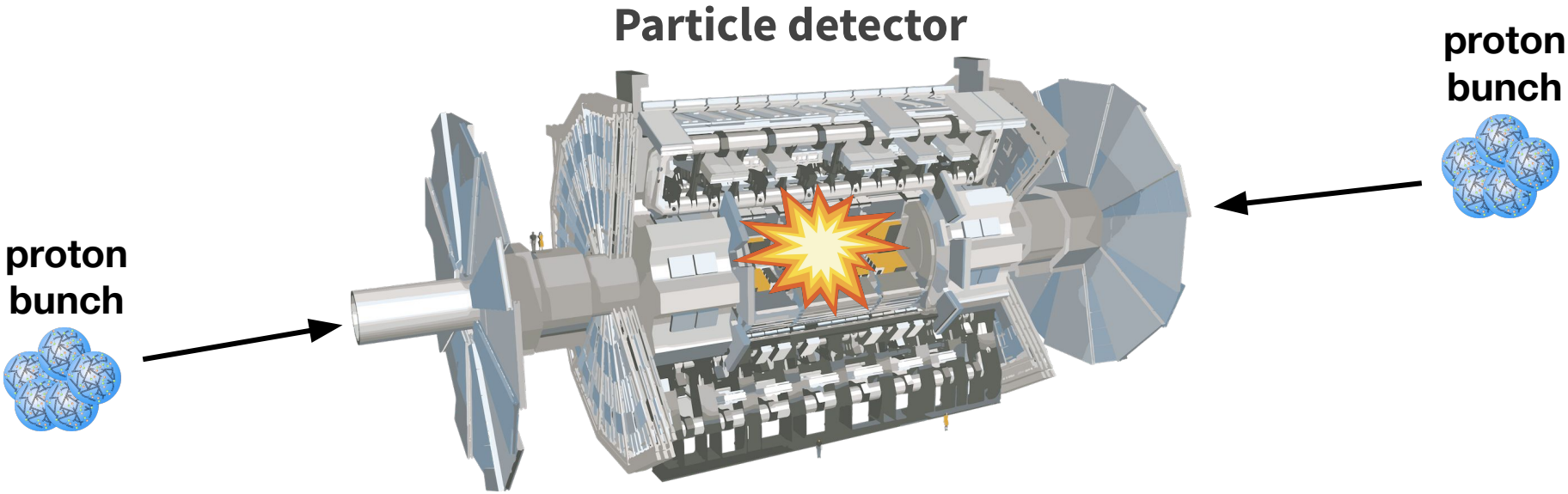
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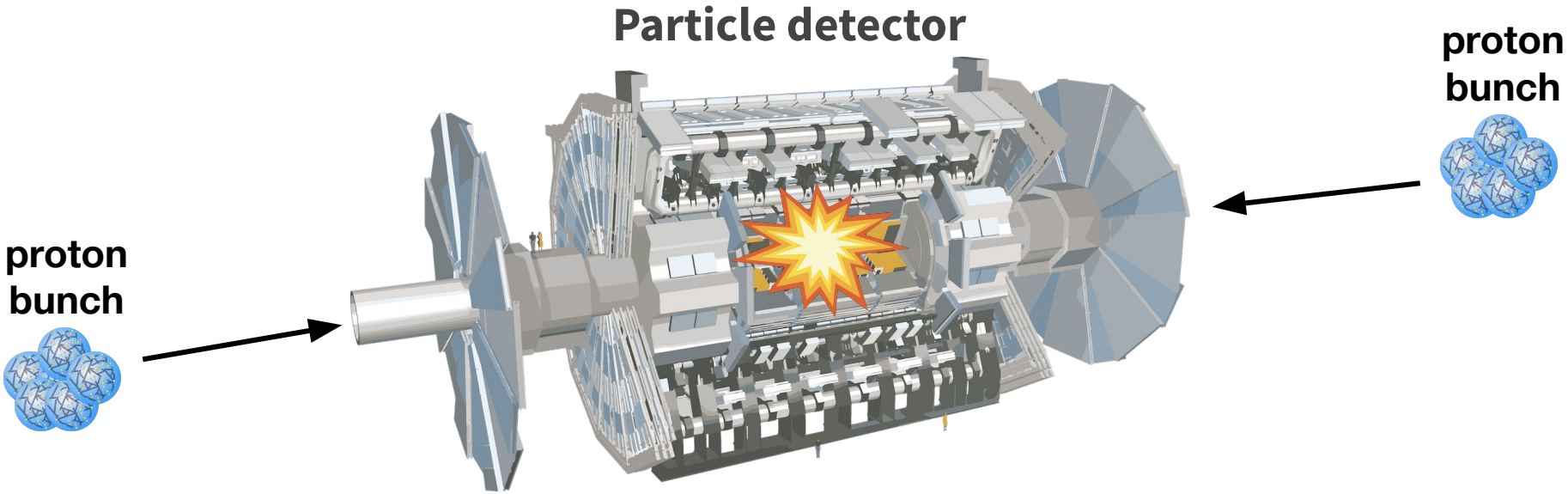
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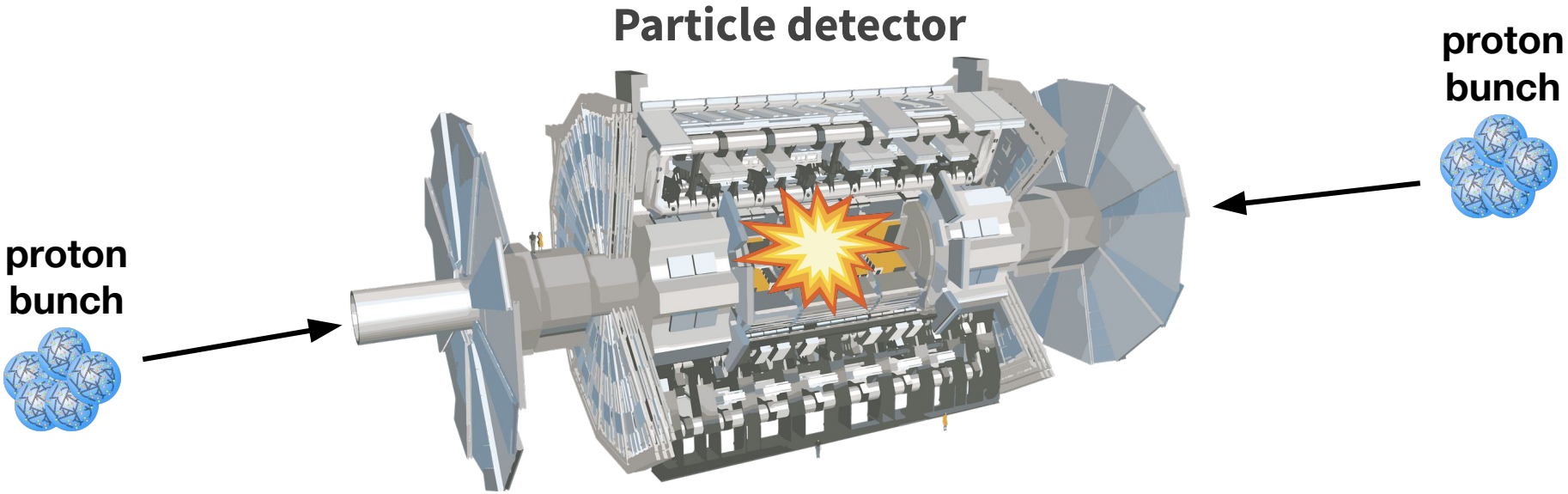


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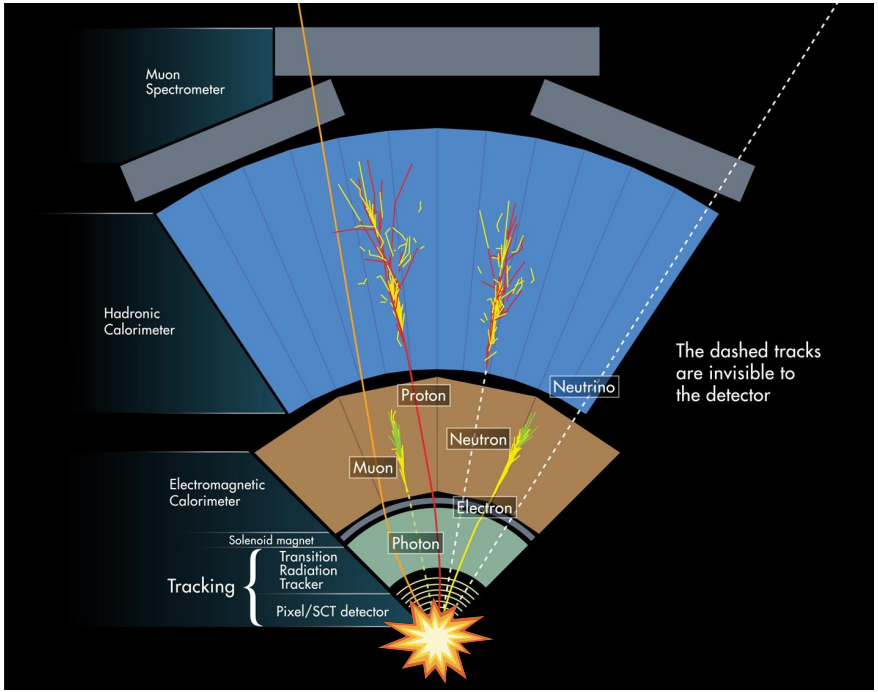


Question: **Why do we collide "bunches" of protons? Why not just collide 2 protons?**

Answer: Very low chance of interaction with two protons. Collide **bunches of protons (~100 billion each!)**
Pro: **More likely to produce our physics processes of interest**
Con: **Have to deal with lots of extra interactions, making environment "messy".**

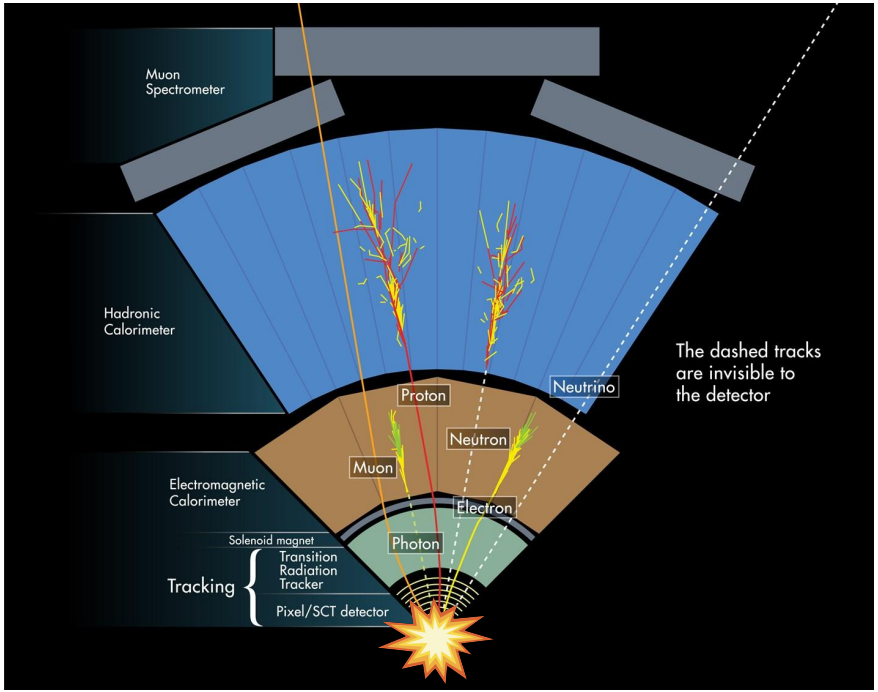
The ATLAS detector

- Different **layers** detect different **particles**



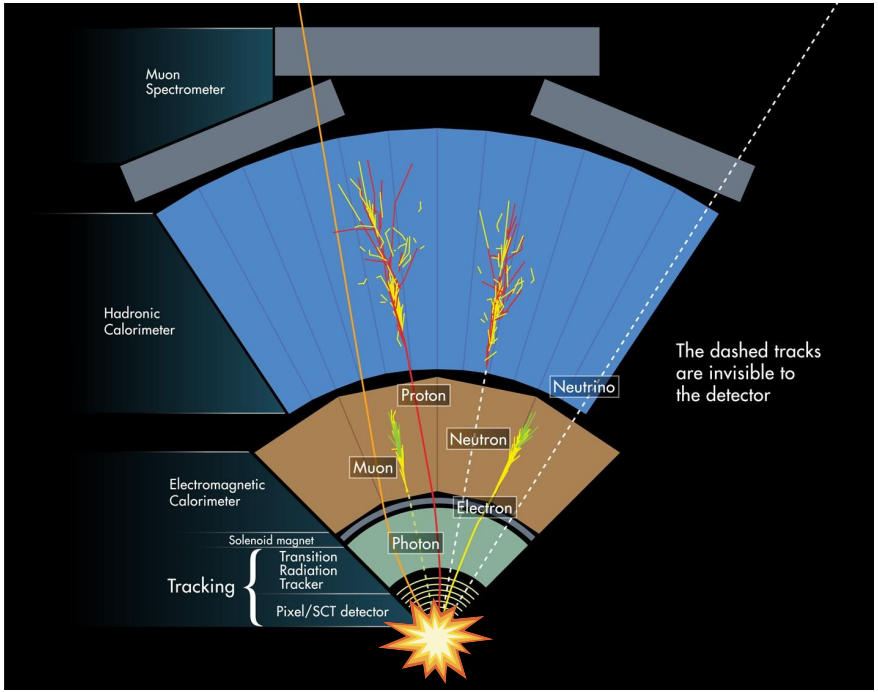
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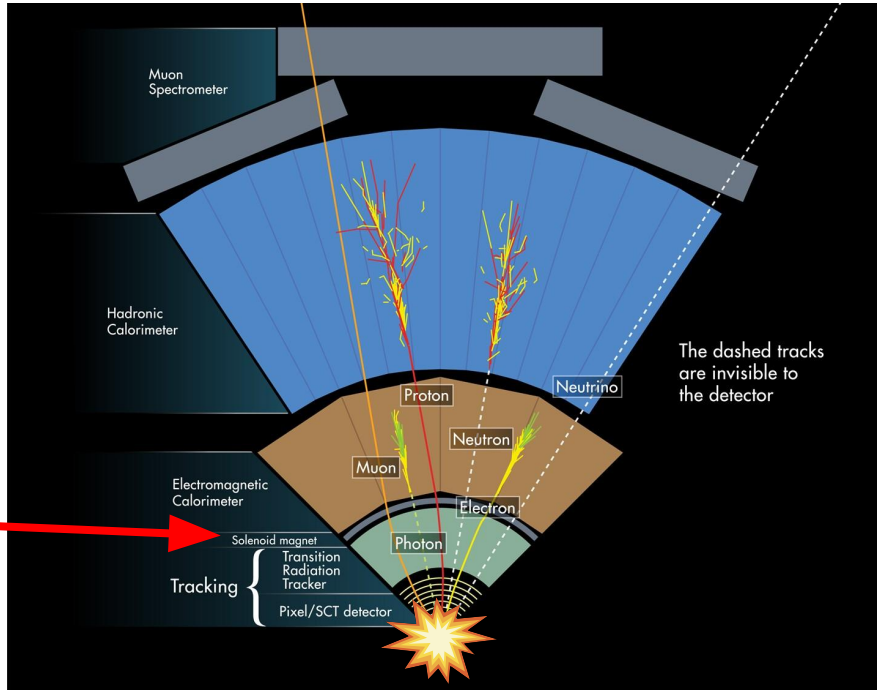
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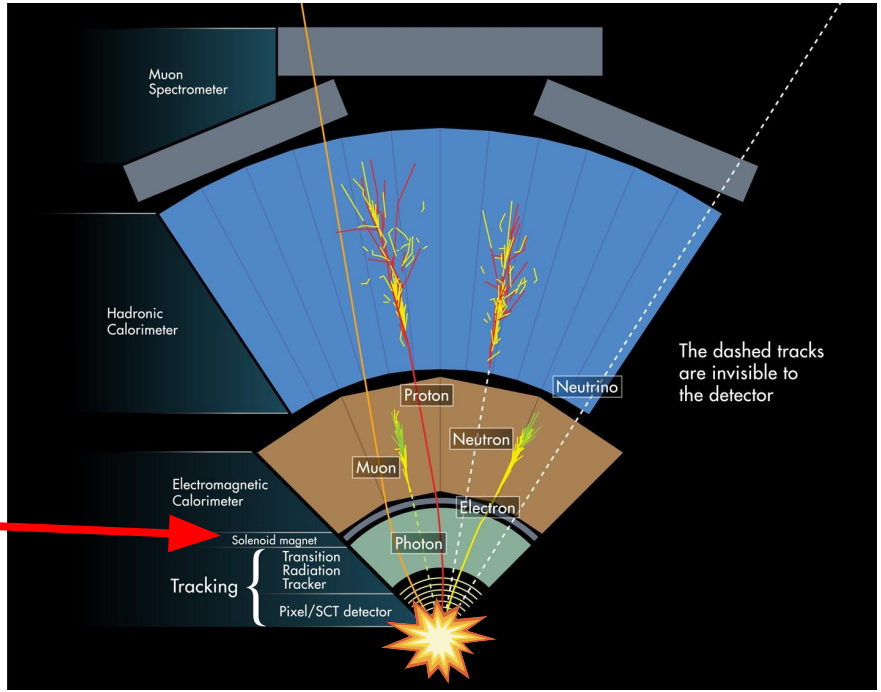


The ATLAS detector

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Question: Why does ATLAS have magnets?

Answer: To bend charged particles, use **Lorentz Force law** to measure **momentum**.



Part I summary

- We study **particle physics** to answer **fundamental questions** about the universe.
- Particle **accelerators** are used to create **collisions**, particle **detectors** are used to **study what comes out of them**.
- We use different particle detector **technologies** to study different **particles**.

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Any questions before we move to Part II?



Next section

Brookhaven National Laboratory

Part I: Why and how we study particle physics

Part II: Detecting particles with silicon

Silicon

- Element, atomic number **14**
- Common in the **Earth's crust**, **rocks**, even **beach sand**! (Through SiO_2)
- Can **refine** SiO_2 to create more pure **Silicon**
- Defined as a "**metalloid**": Has traits of both **insulators** and **conductors**.



periodictable.com

Processing Silicon



Processing Silicon



Refine



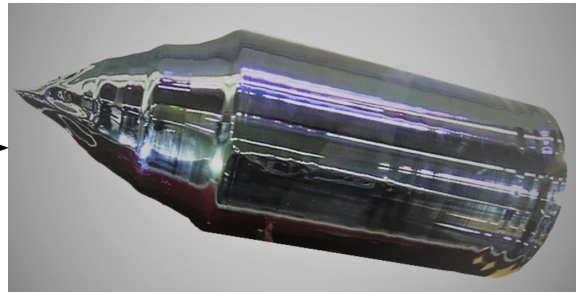
Processing Silicon



Refine



Melt + form



Processing Silicon



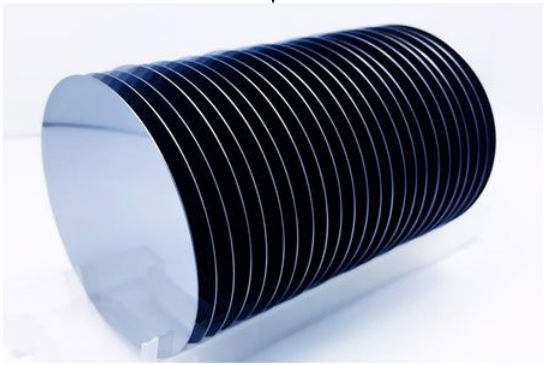
Refine



Melt + form



Slice



Processing Silicon



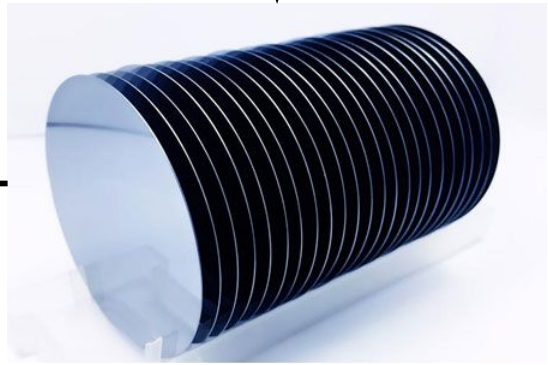
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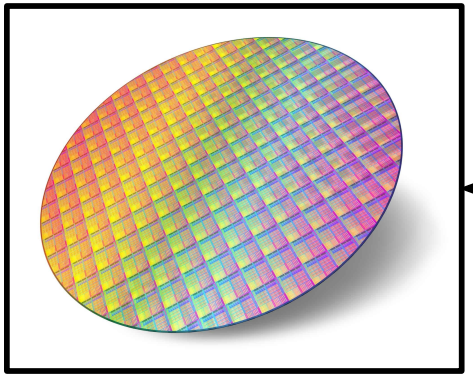
Melt + form



Slice



Etch the design



Processing Silicon



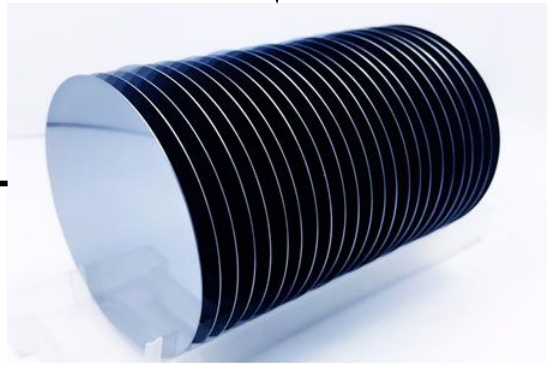
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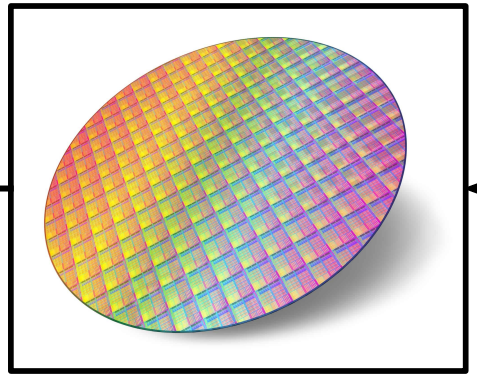
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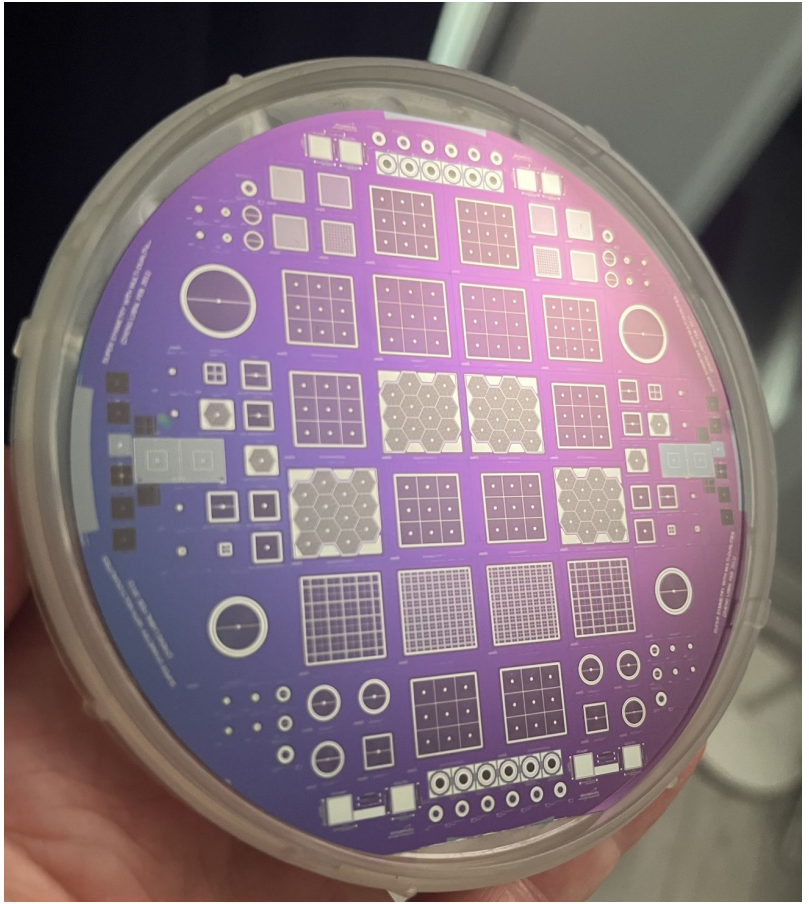


Final product!



Processing Silicon

Example wafers from BNL!

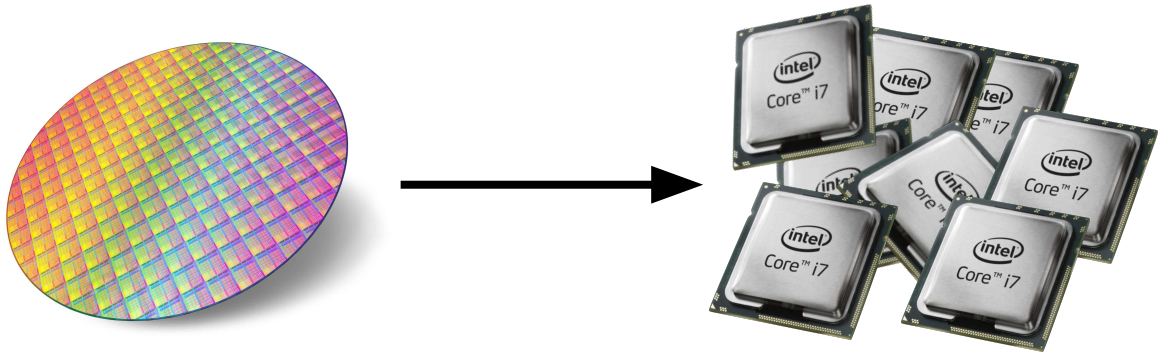


Applications

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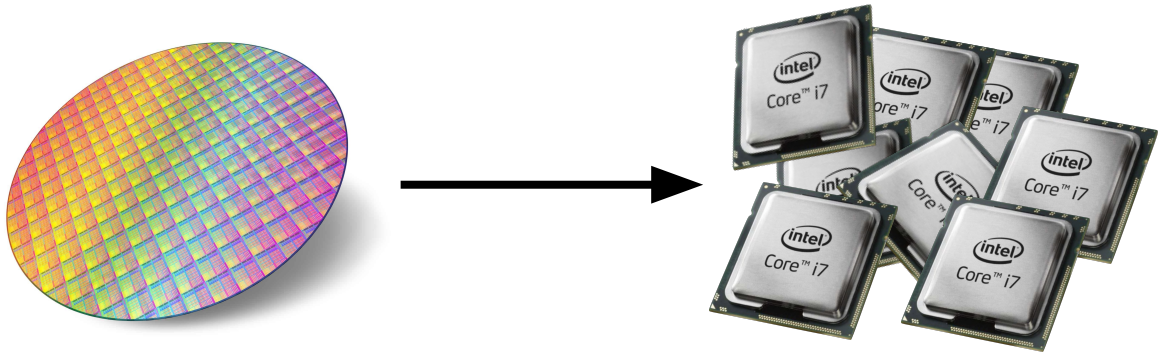
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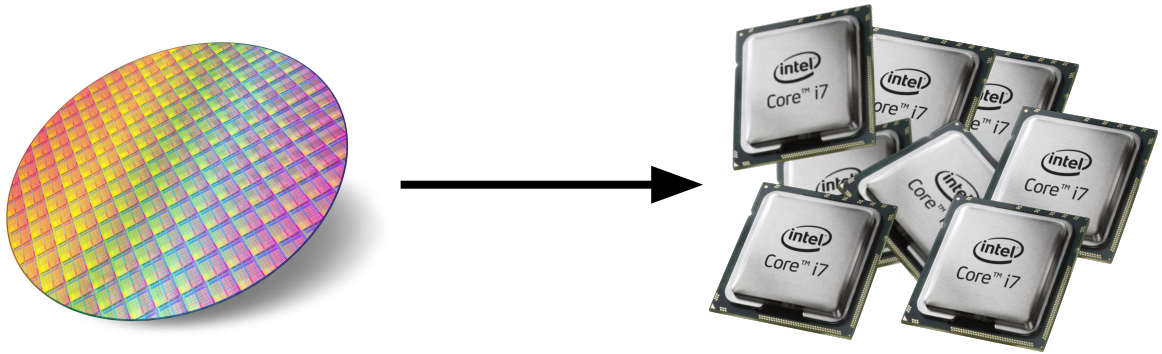
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- Very deep experience in **industry** and **academia** in how to **process silicon for custom applications**.

Applications

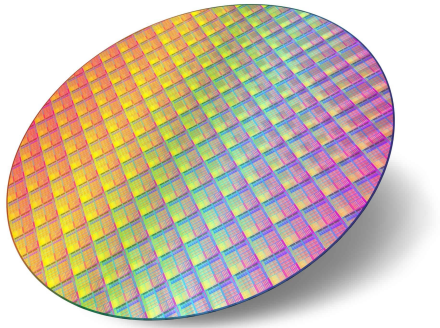
- Silicon is a staple in many **industrial** and **scientific** applications, e.g. **microelectronics**
- Widely used for **computer chips**, for example Central processing units (**CPUs**) and Graphic processing units (**GPUs**):



- Very deep experience in **industry** and **academia** in how to **process silicon for custom applications**.
- **How can we use silicon for particle physics?**

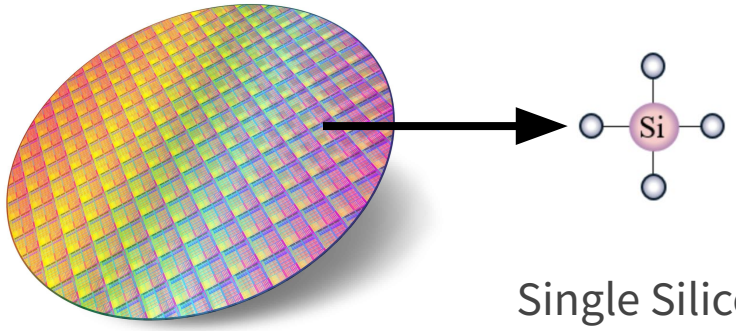
The lattice

- Zooming into the completed silicon wafer



The lattice

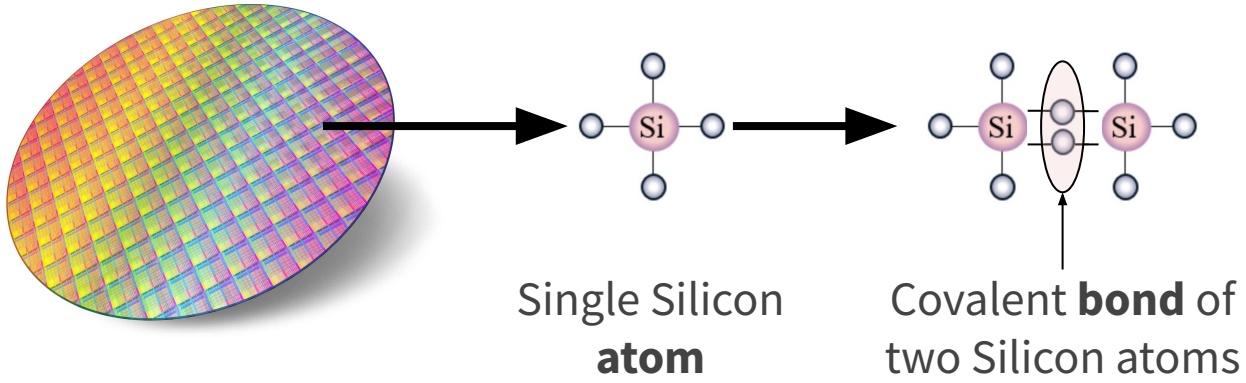
- Zooming into the completed silicon wafer
- Atomic number 14, so **4 valence electrons**



Single Silicon
atom

The lattice

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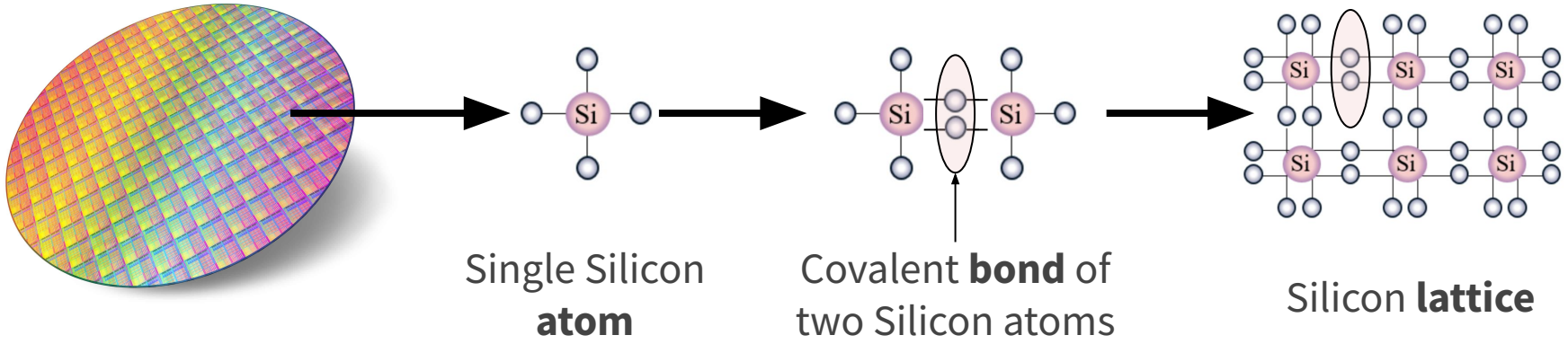


Single Silicon **atom**

Covalent **bond** of two Silicon atoms

The lattice

- Zooming into the completed silicon wafer
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Single Silicon **atom**

Covalent **bond** of two Silicon atoms

Silicon **lattice**

shindengen.com

Doping

- On its own, a silicon lattice not particularly useful for physics - **does not conduct electricity!** (just have a bunch of **stable covalent bonds**)

Doping

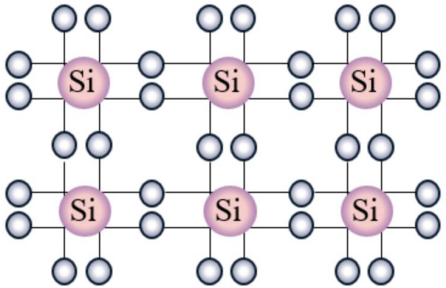
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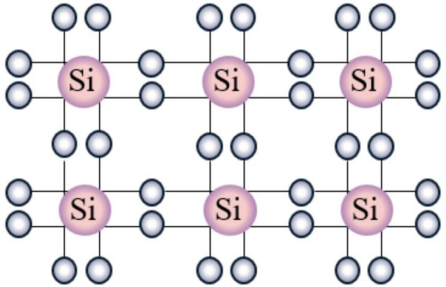
Silicon **lattice**

Doping

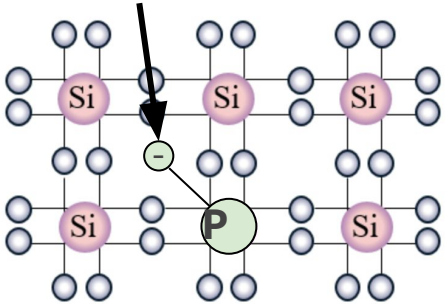
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Extra, free electron



Silicon **lattice**

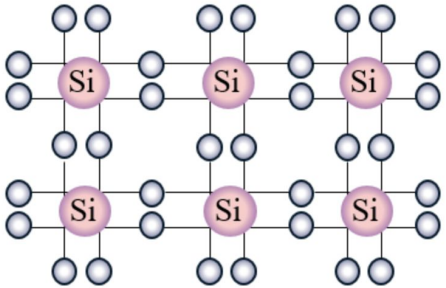


“**n-type**” doping

Doping

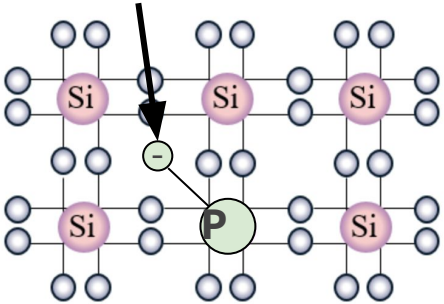
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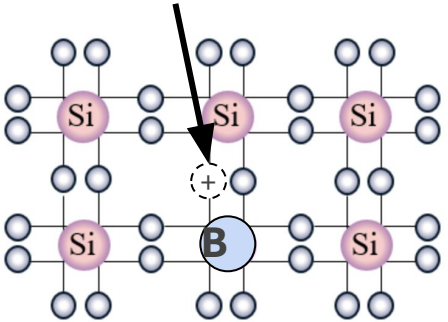
Silicon **lattice**

Extra, free electron



"**n-type**" doping

Hole: Absence of electron

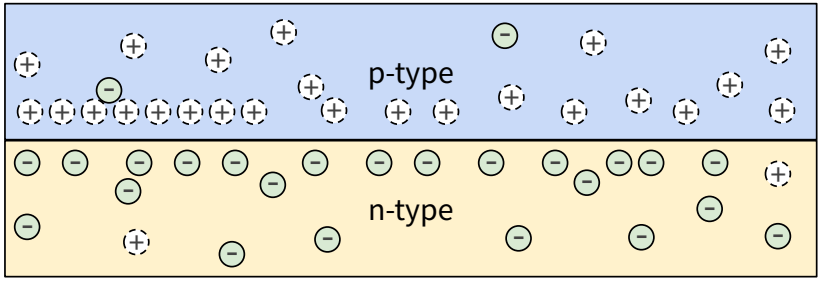


"**p-type**" doping

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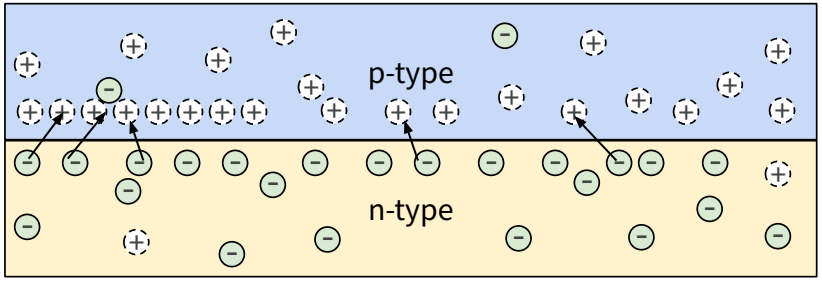
PN junction

- When p and n-type placed together,



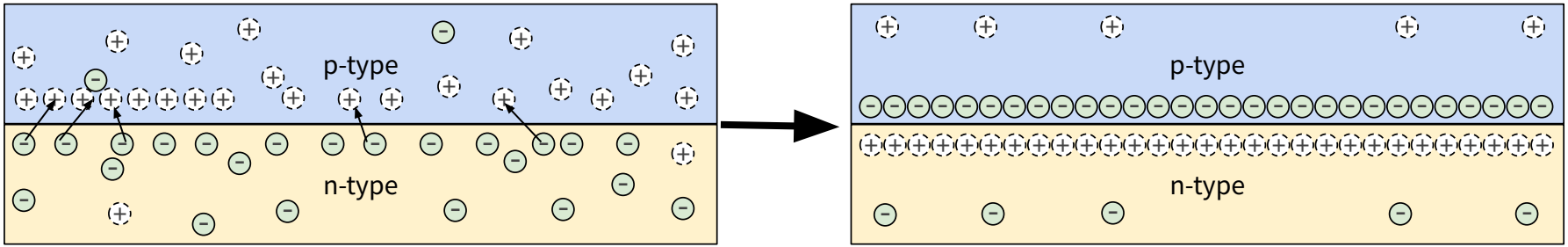
PN junction

- When p and n-type placed together, **diffusion** occurs,



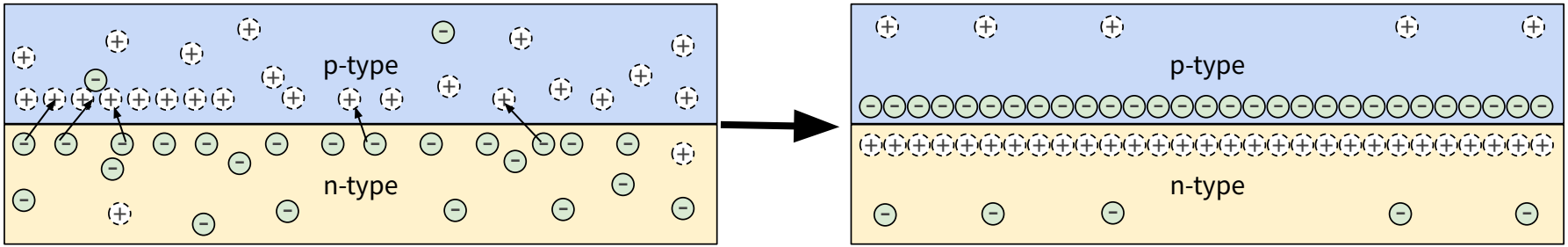
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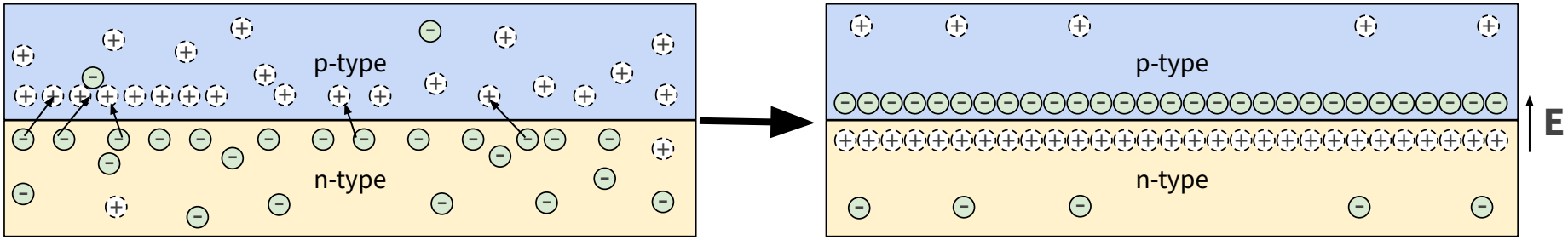


Effectively negative and positive charges.

Question: Any sort of **“field”** created between them?

PN junction

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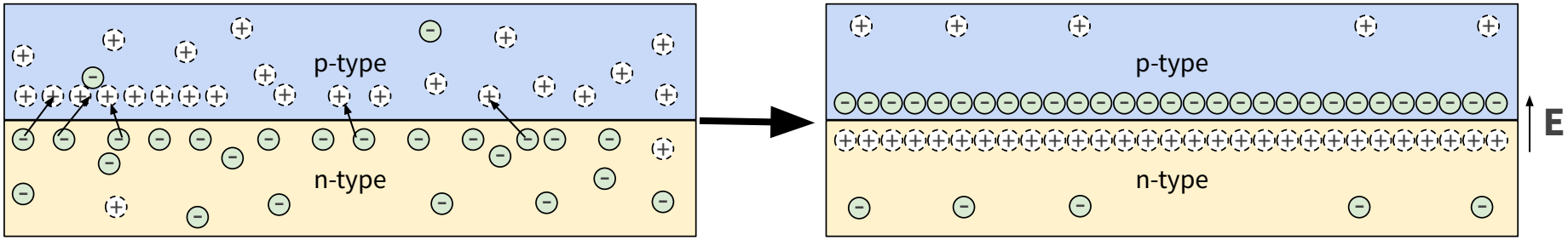


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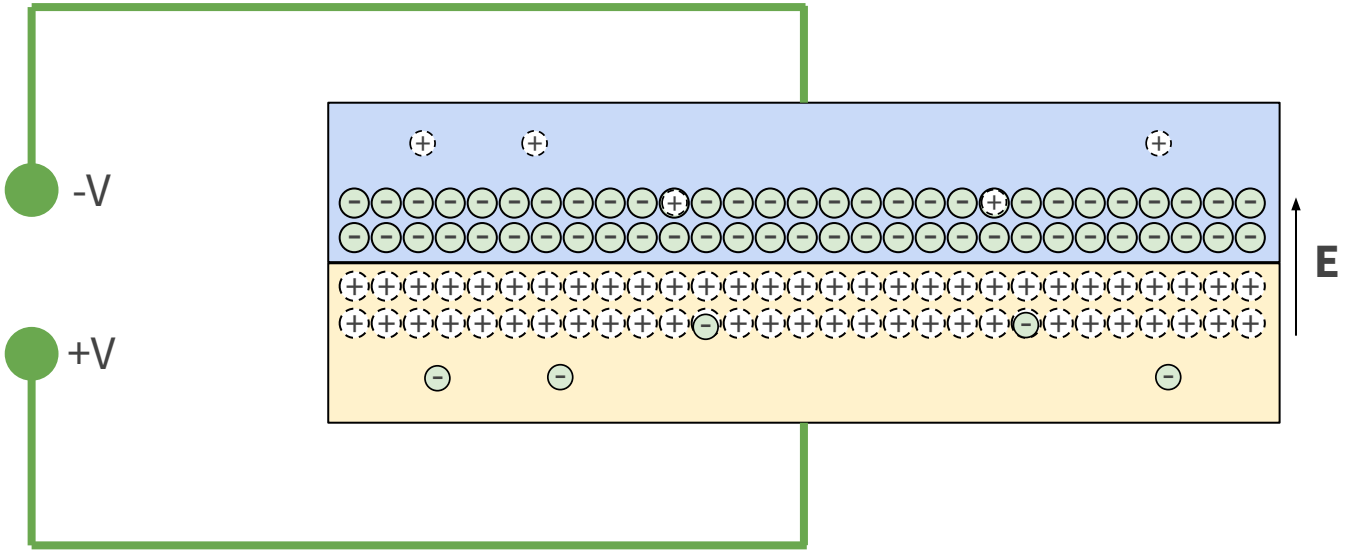
→ Prevents further movement of electrons or holes across the boundary - call this the **depletion region** (no "free" electrons or holes)

Bias voltage

- What happens if we apply a **voltage** across the PN junction?

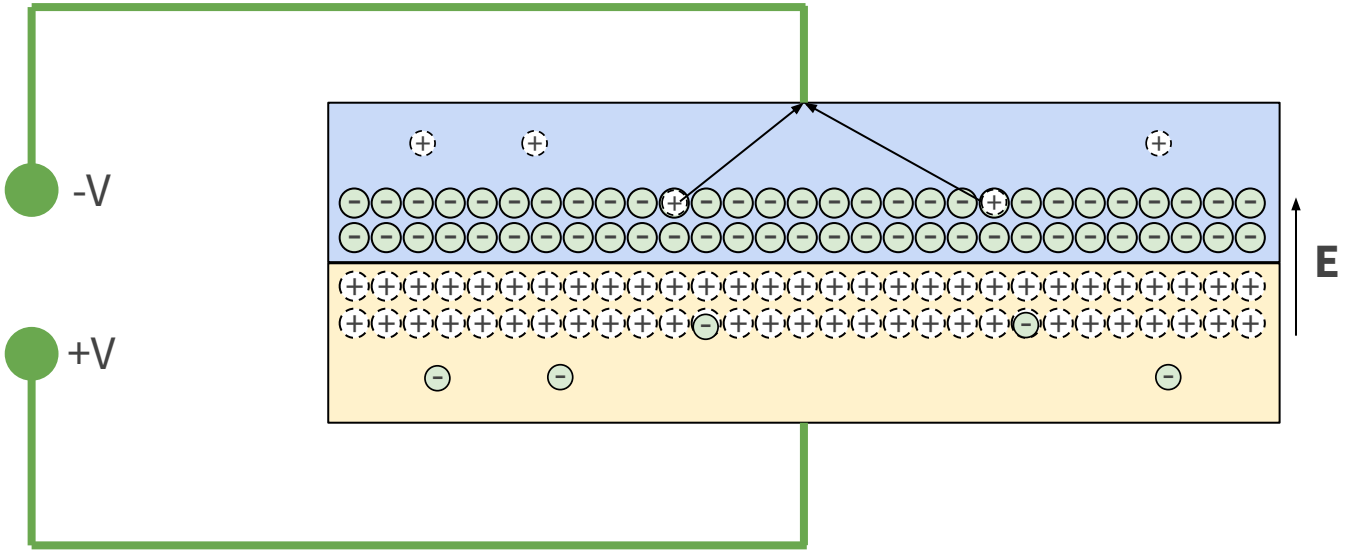
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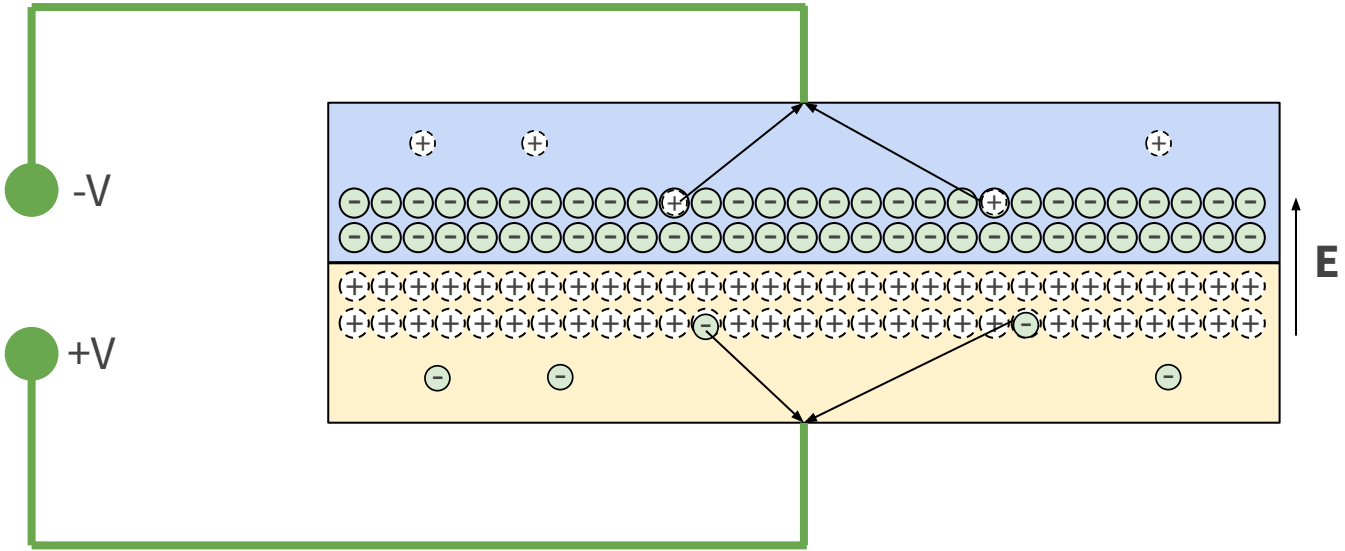
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- Remove holes from top, leaves negative charge behind (-).

Bias voltage

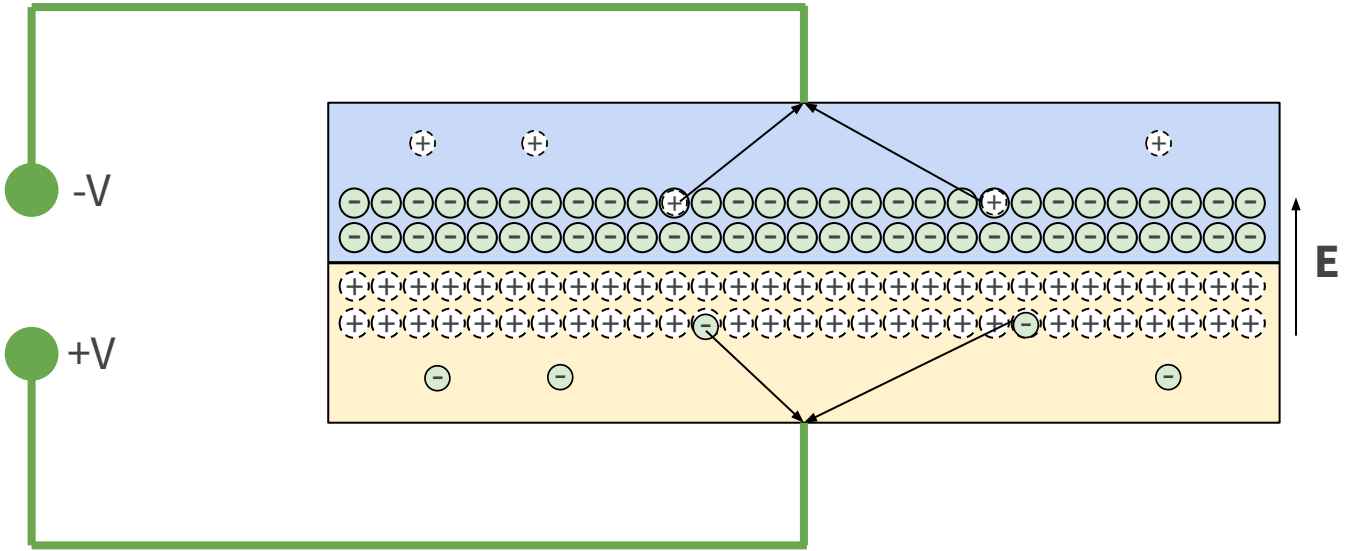
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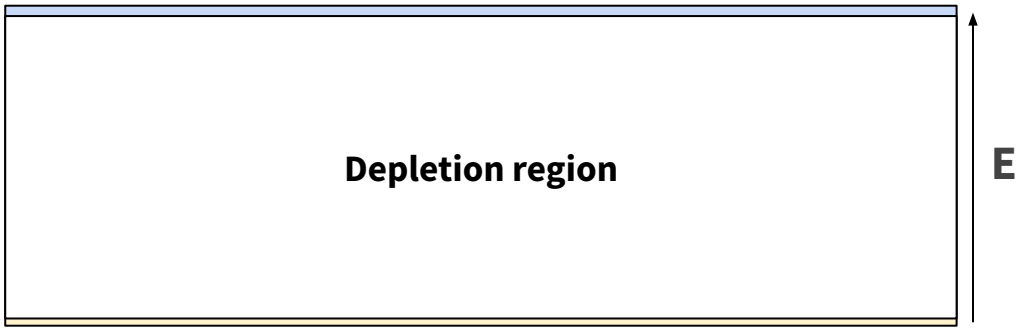
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- **Increase in the depletion region**

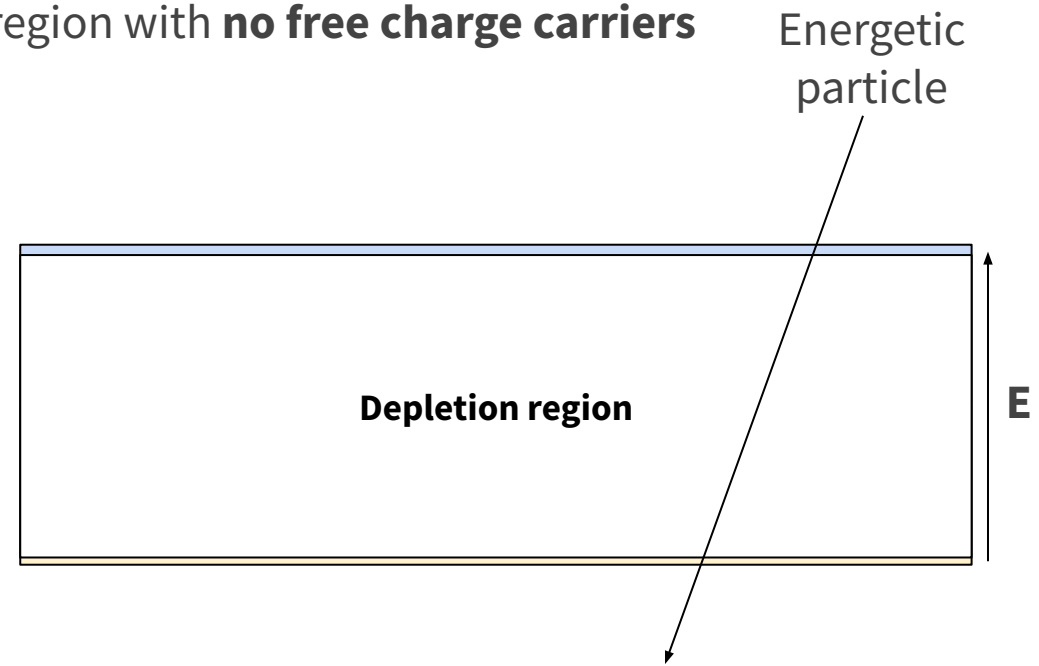
Ionization and collection

- Now have a region with **no free charge carriers**



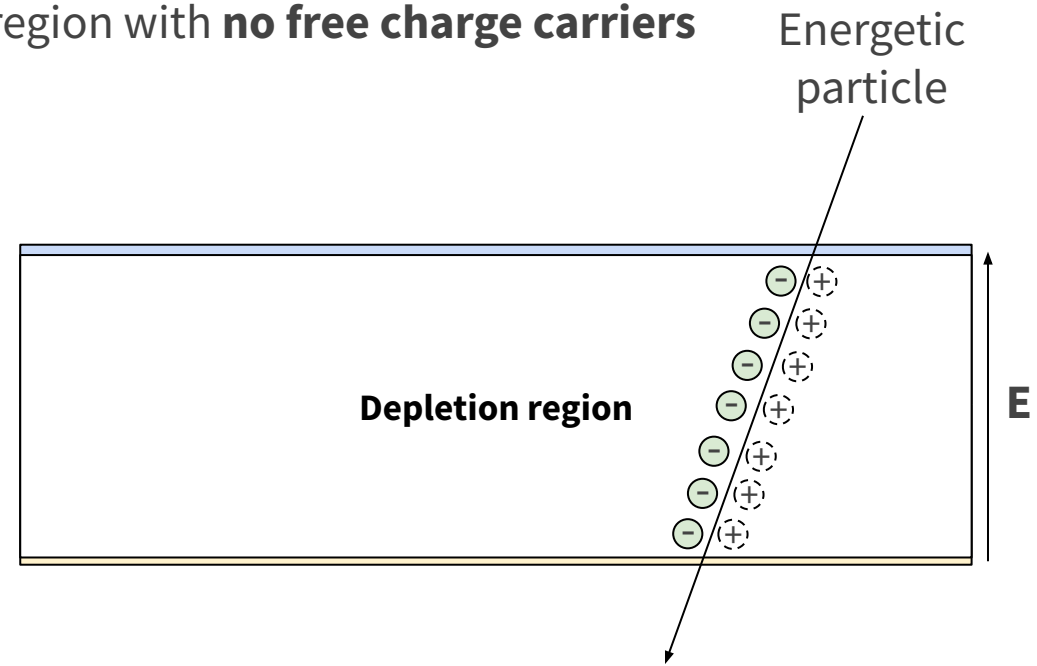
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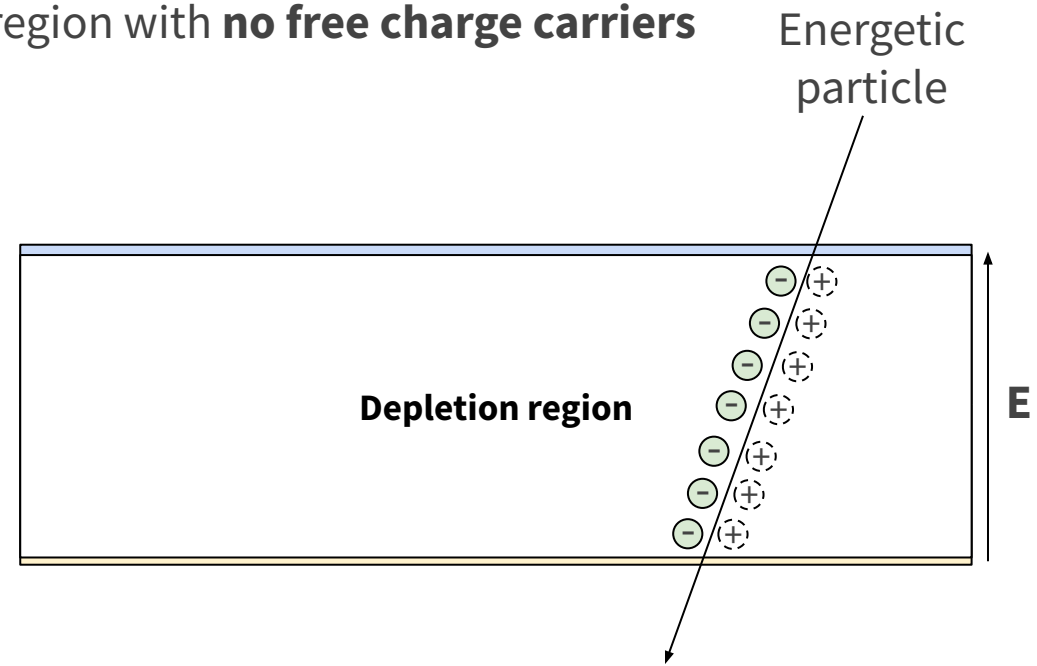
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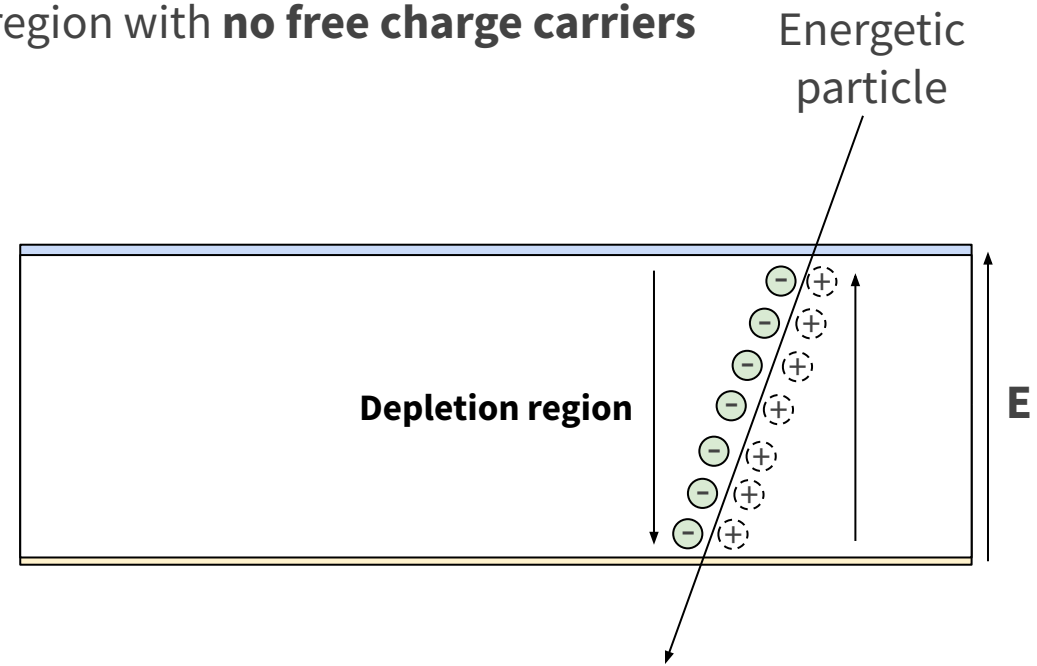
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Ionization and collection

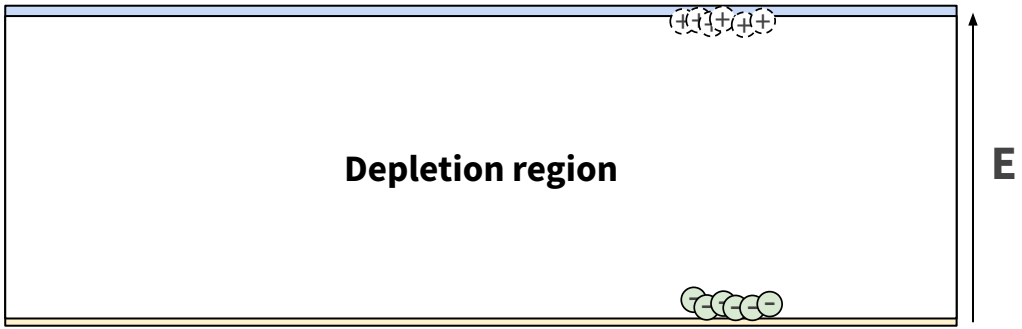
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Question: What will happen to these new **free** electrons and holes?
Answer: They will experience an **electric force** and accelerate in either direction

Ionization and collection

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This means we can **collect charge** on either side of the device.

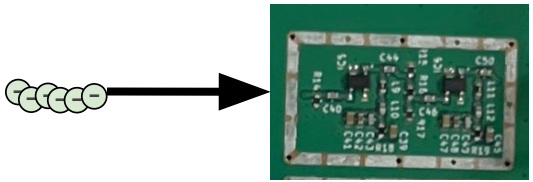
Waveforms

- When the charges move, this creates **electric current** (charge as a function of time)



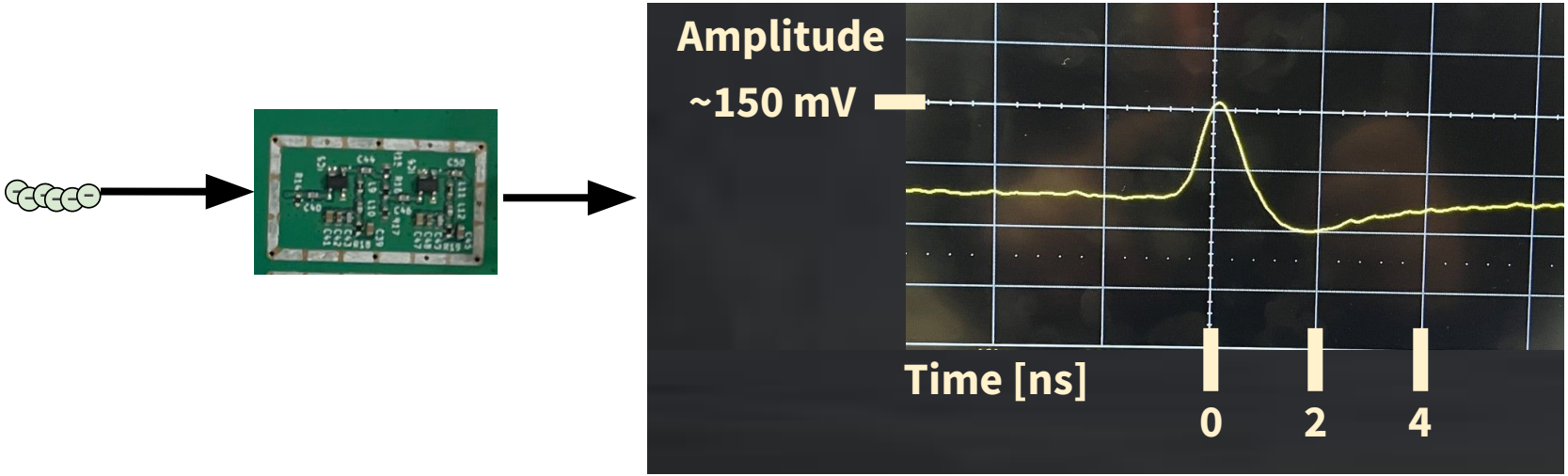
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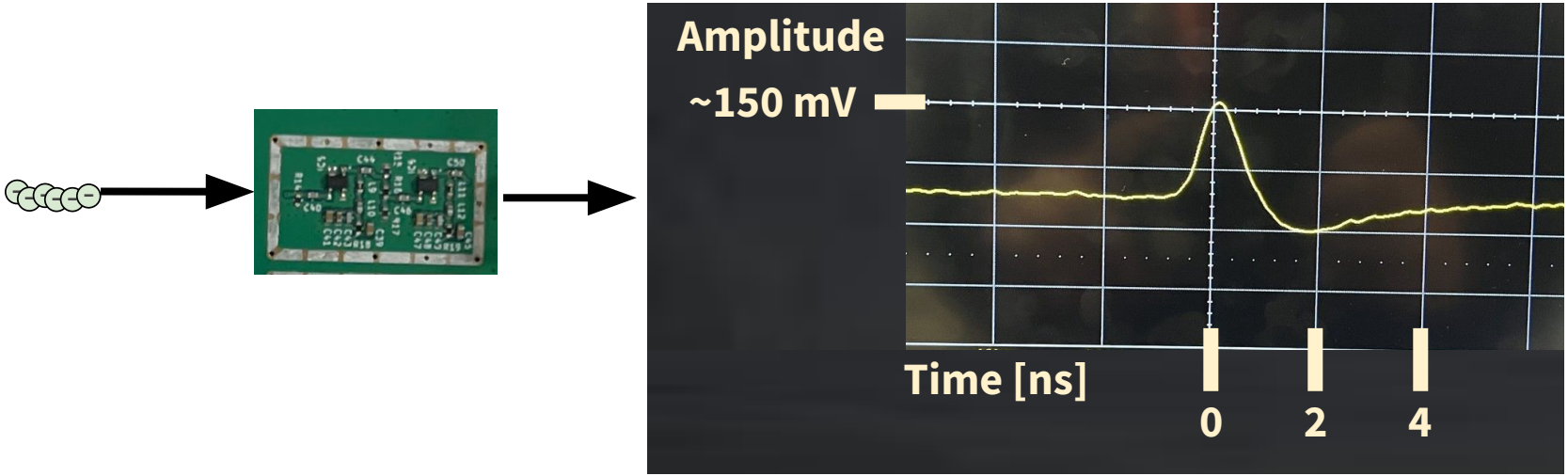
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Waveforms

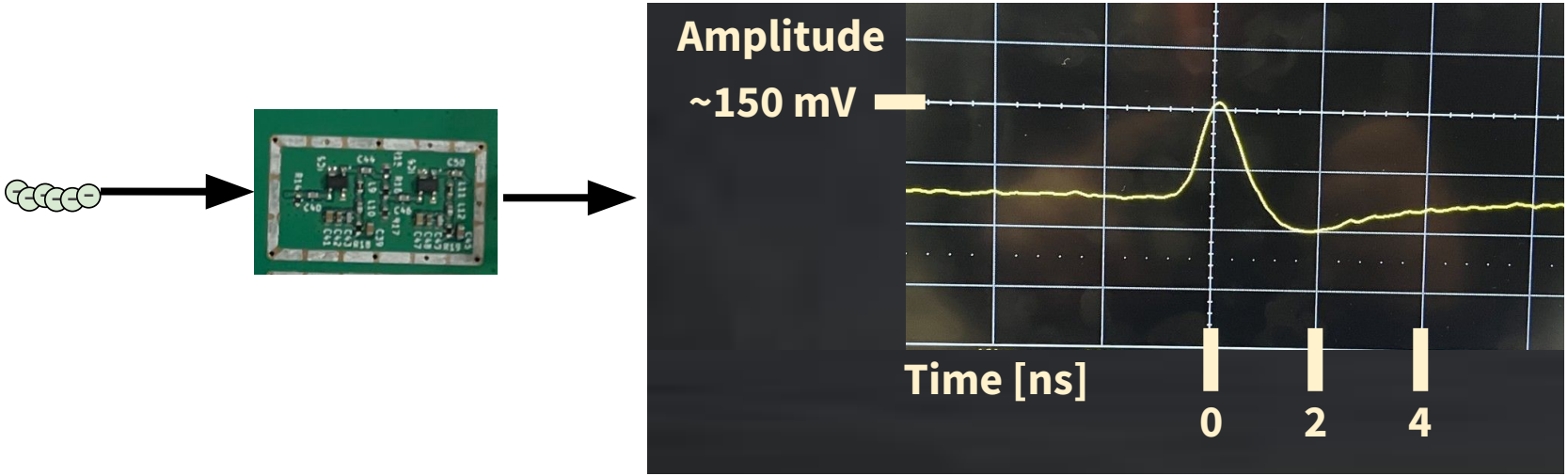
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Signal lasts for **~ a billionth of a second**!!

Waveforms

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- Gives us a **waveform** (example I took at **BNL**):



Signal lasts for **~ a billionth of a second**!!

- This means if we see a **waveform**, it might be a **signal from a particle**!

Signal analysis

- **What can we do with this signal waveform?**

Signal analysis

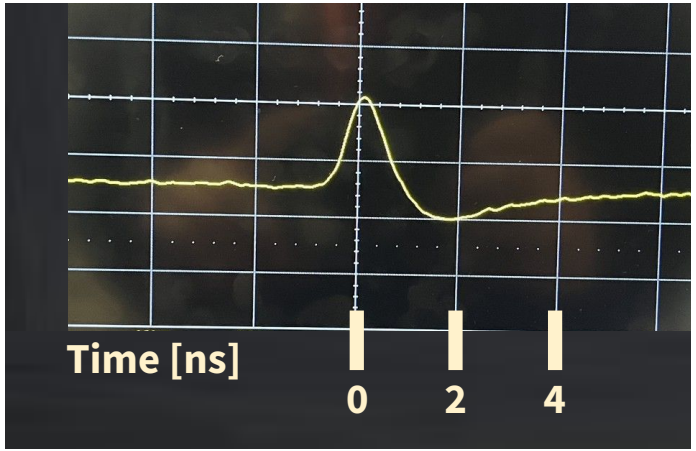
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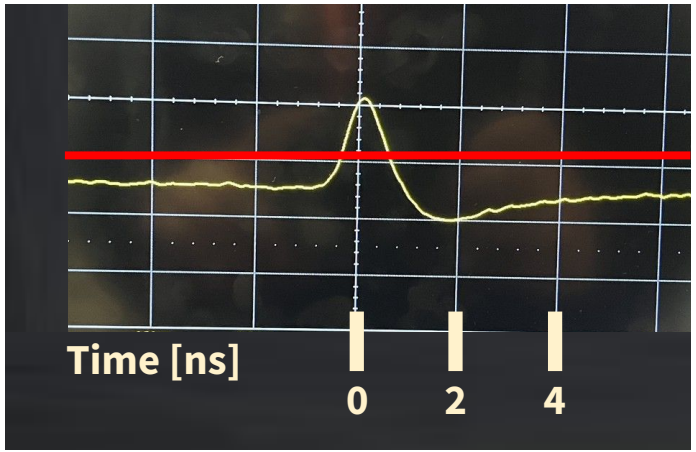
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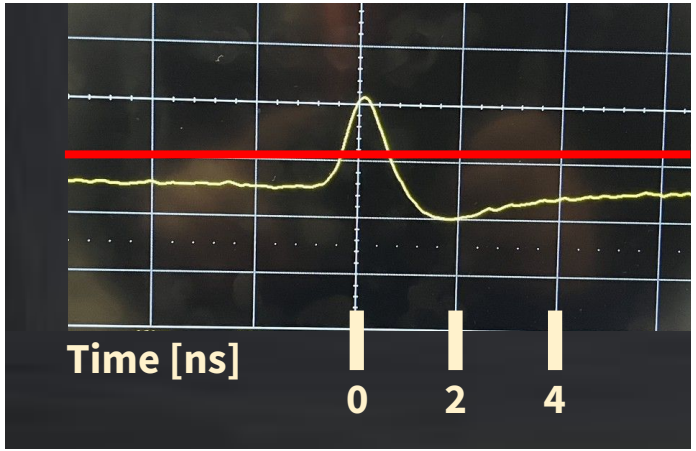
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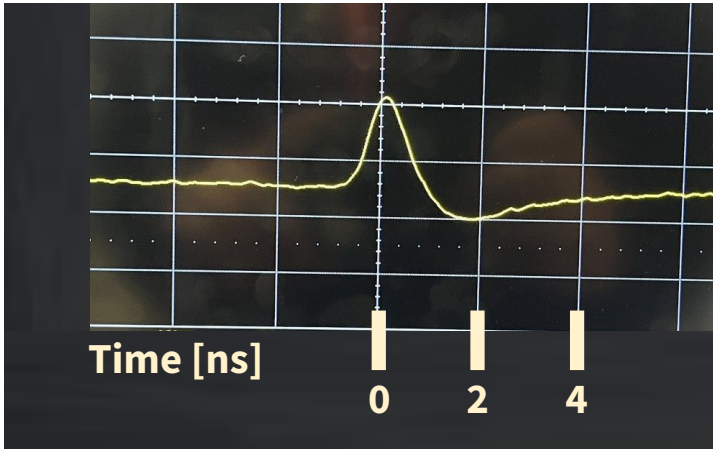
Classify as a **hit**

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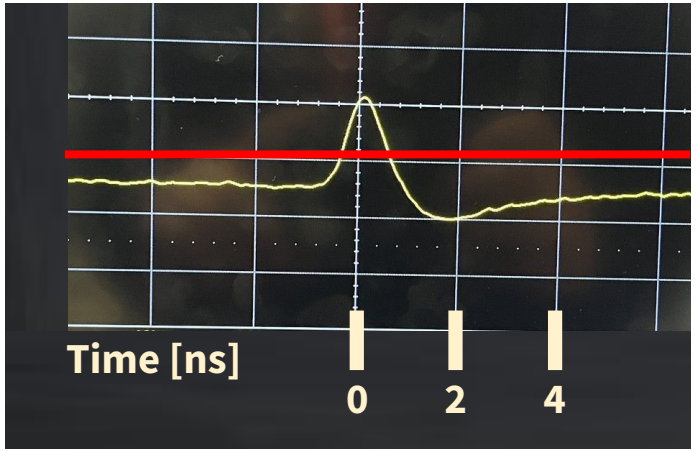


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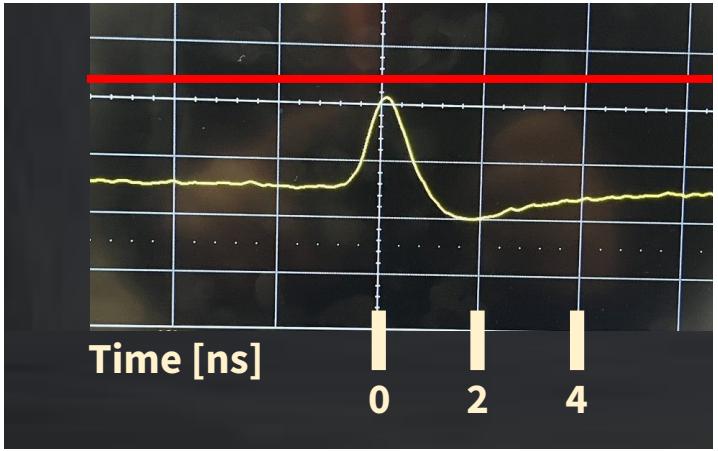


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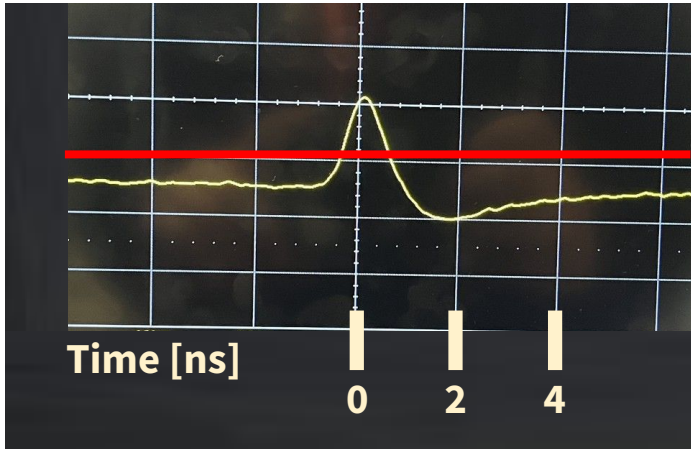


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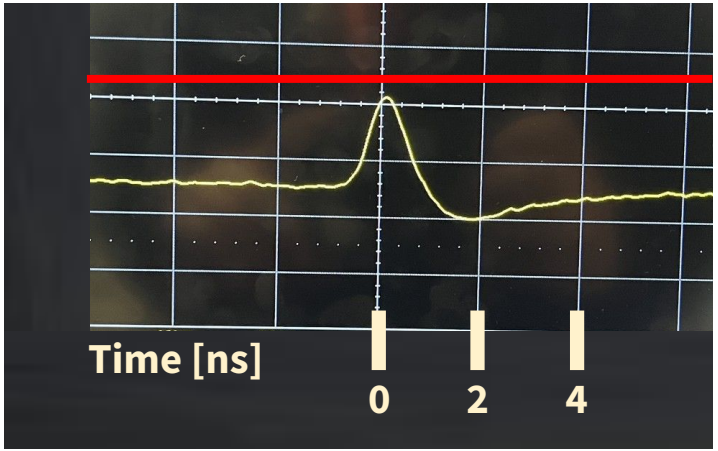


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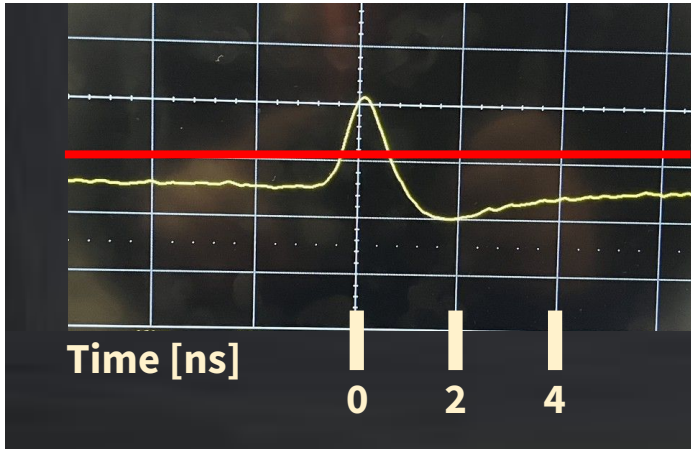
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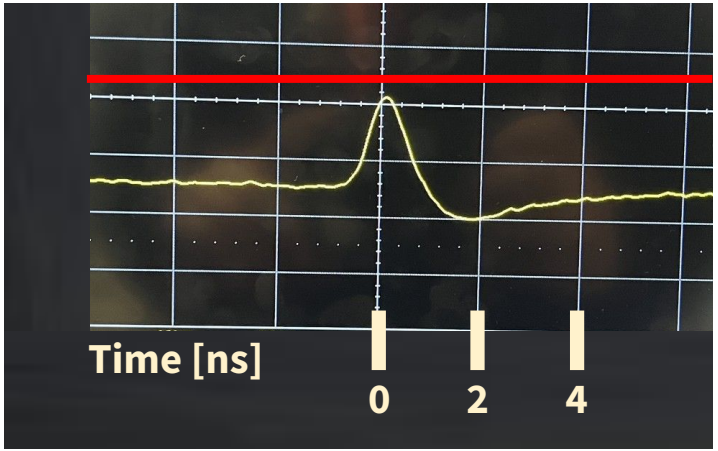
Classify as **not a hit**

Signal analysis

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Classify as a **hit**

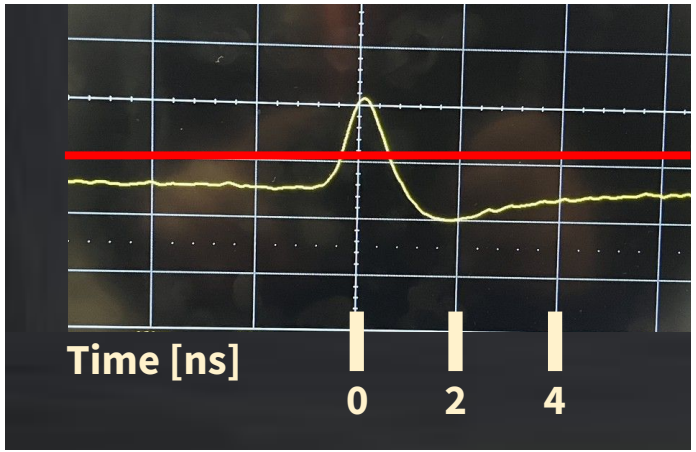


Classify as **not a hit**

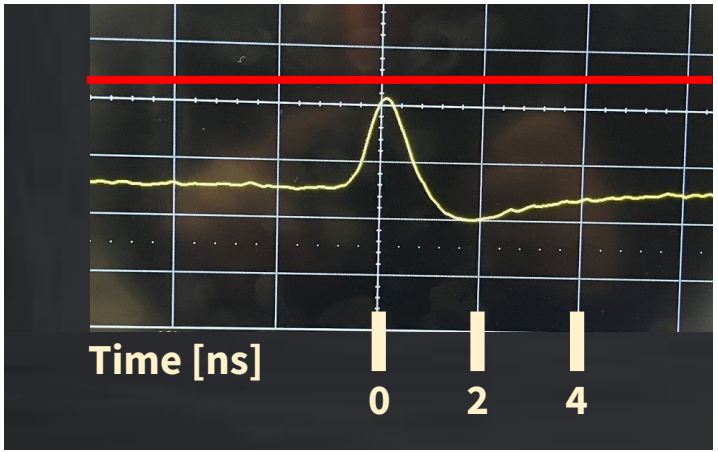
- **Question:** Why don't we just set the height **very low** so that everything we see is a hit?

Signal analysis

- **What can we do with this signal waveform?**
- One example: **Binary classification**
- If waveform goes above a certain height, consider this a **hit**



Classify as a **hit**



Classify as **not a hit**

- **Question:** Why don't we just set the height **very low** so that everything we see is a hit?
- **Answer:** Want to avoid **noise** (very common from **silicon** and **electronics**).

Example

- **Question:** How can this be useful for particle physics?

Example

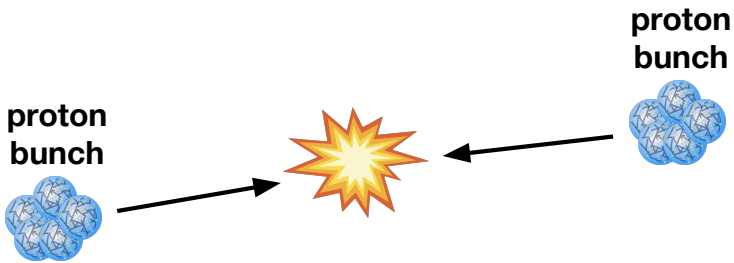
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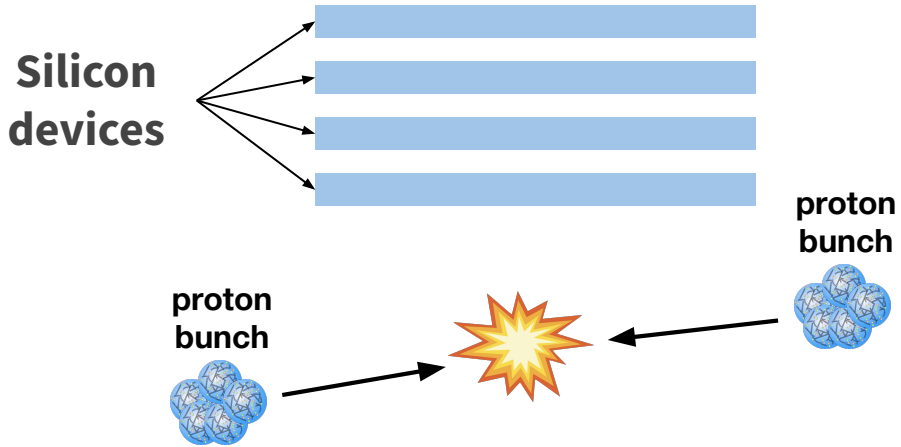
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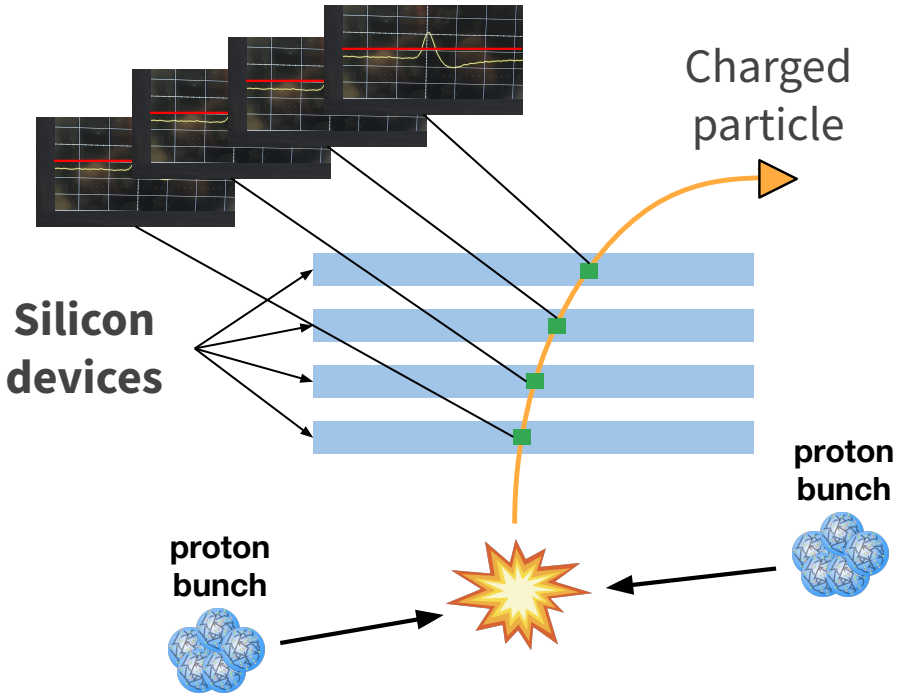
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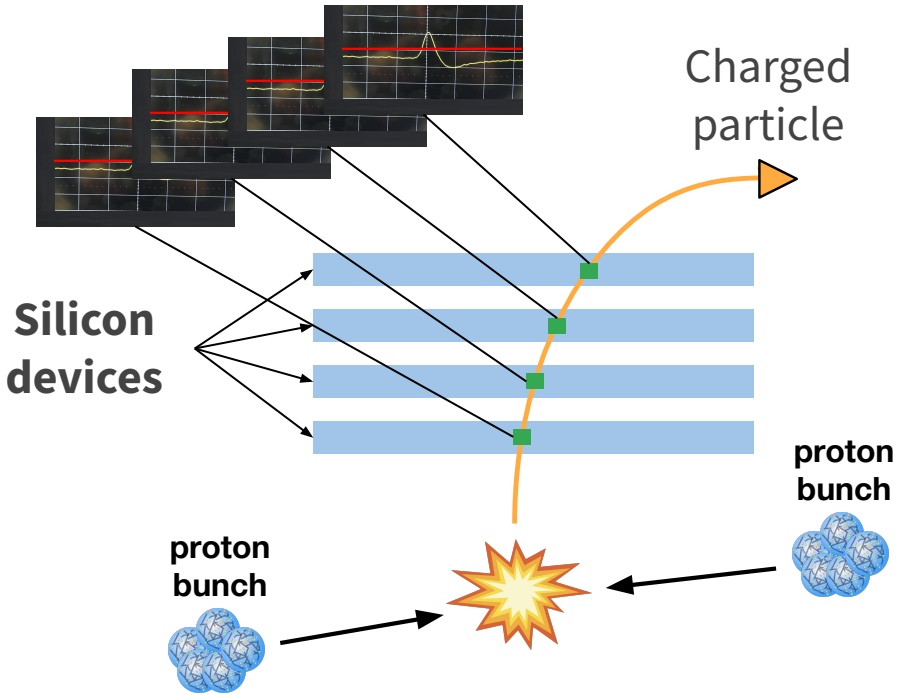
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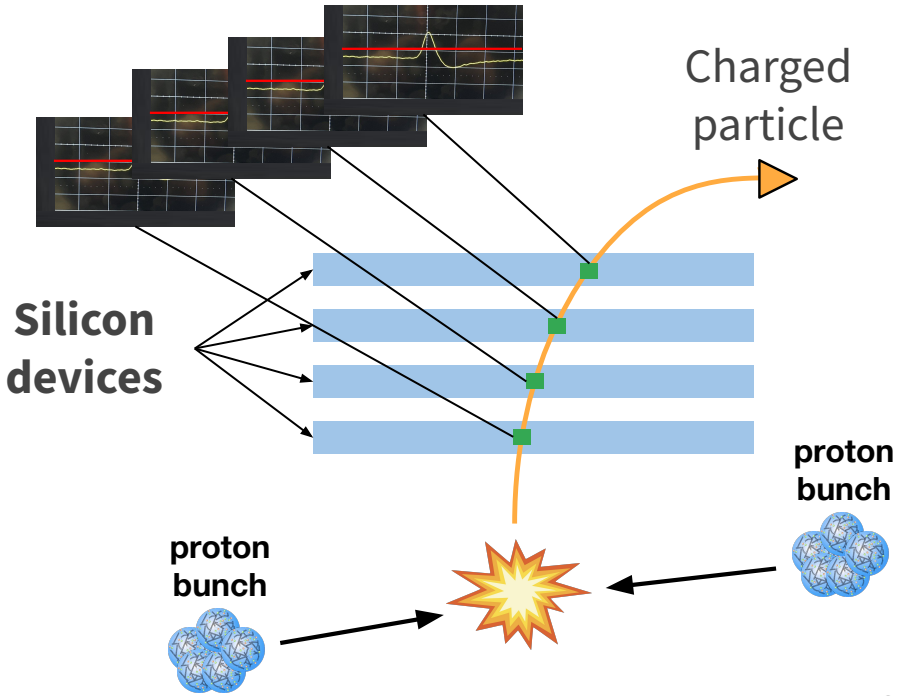


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- All with the goal of answering our **original physics questions**!



Research

- There is a lot of **very cool research** ongoing for silicon detectors!

Research

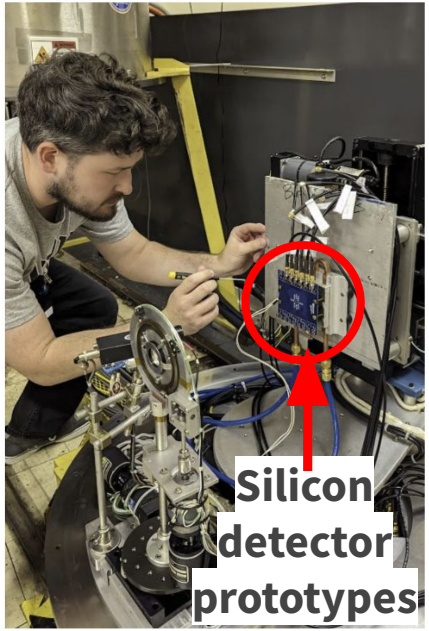
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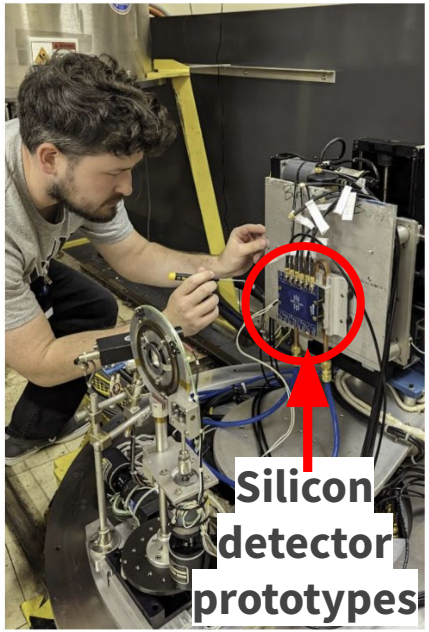




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Something even better than silicon (?)



Summary



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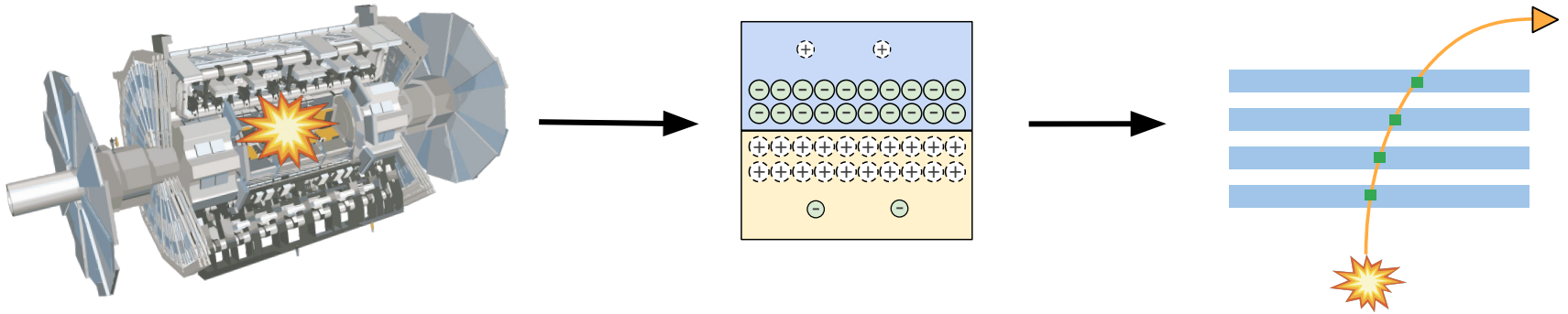
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- Use measurements to try and answer **fundamental questions about the universe**
- Lots of **active research** in silicon detectors. **Very fun place to get involved**!
- Feel free to **reach out** to me if you have questions in the future!
  → ***abraham.tishelman.charny@cern.ch***




Thank you for your attention!